

Coursework 2: BOTSw3 Incident Analysis Report

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Module: COMP3010HK Security Operations and Incident Management

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Contents

1. [GitHub](#)
2. [Video Presentation](#)
3. [1. Introduction](#)
4. [2. SOC Roles & Incident Handling Reflection](#)
 - [Preventive Controls Analysis](#)
 - [Detection & Response](#)
 - [Conclusion on SOC Visibility](#)
5. [3. Installation & Data Preparation](#)
 - [Installation Evidence](#)
 - [Data Validation](#)
6. [4. Guided Questions Analysis](#)
 - [Question 1: Identifying IAM Users](#)
 - [Question 2: Multi-Factor Authentication Check](#)
 - [Question 3: Hardware Inventory](#)
 - [Question 4-6: S3 bucket](#)
 - [Question 7: Impact Assessment](#)
 - [Question 8: FQDN](#)
7. [5. Conclusion, References and Presentation](#)
8. [Reference](#)
9. [Appendix A - Cryptojacking](#)
10. [Appendix B - Student Declaration of AI Tool Use](#)
11. [Appendix C - GitHub Repository Availability & Incident Report](#)
 - [1. Incident Summary](#)
 - [2. Impact on Submission and Incident Response & Verification](#)
 - [3. Evidence of the Github work](#)

GitHub

If the link returns a 404 error, please refer to Appendix C – GitHub Repository Availability & Incident Report for details.

https://github.com/10957621/COMP3010HK_ASSESSMENT_CW1-CW2.git

COMP3010HK_ASSESSMENT_CW1-CW2

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Code

10957621

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142 Commits

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CW2	Add files via upload	3 hours ago
README.md	Initial commit	last month

README

COMP3010HK_ASSESSMENT_CW1-CW2

Coursework for COMP3010HK Security Operations & Incident Management - PCAP analysis (CW1) and BOTSv3 Incident Analysis and Presentation(CW2)

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CodeIssuesPull requestsActionsProjectsWikiSecurityInsightsSettings

Files

main

Go to file

CW1

CW2

Installation & Data Preparation

Quiz

README.MD

Student Declaration of AI Too...

preview.png

youtubepreview.png

README.md

COMP3010HK_ASSESSMENT_CW1-CW2 / CW2

10957621 Add files via uploadfb0ba2b · 3 hours agoHistory

Name	Last commit message	Last commit da...
..		
Installation & Data Preparation	Delete CW2/Installation & Data Preparation/0...	last week
Quiz	Add files via upload	yesterday
README.MD	Update README.MD	3 hours ago
Student Declaration of AI Tool use i...	Add files via upload	3 hours ago
preview.png	Add files via upload	3 hours ago
youtubepreview.png	Add files via upload	3 hours ago

Video Presentation

<https://www.youtube.com/watch?v=SM90B3rc2Ik>

YouTubeJP

搜尋

COMP3010HK_ASSESSMENT_CW1-CW2

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2 Public

CodeIssuesPull requestsActionsProjectsWikiSecurityInsightsSettings

FilesmainCOMP3010HK_ASSESSMENT_CW1-CW2 / CW2

10957621 Update README.MD7ba7a38 · 21 minutes

Name	Last commit message	Last s
..		
Installation & Data Preparation	Delete CW2/Installation & Data Preparatio...	22 m
Quiz	Add files via upload	22 m
README.MD	Update README.MD	21 m

README.MD

BOTSv3 Incident Analysis Report

Student ID: [10957621] Module: COMP3010HK Date: February 2026

GitHub

[\[https://github.com/10957621/COMP3010HK_ASSESSMENT_CW1-CW2.git\]](https://github.com/10957621/COMP3010HK_ASSESSMENT_CW1-CW2.git)

Video Presentation

[Click here to watch my video presentation on YouTube](#)

Oracle VM VirtualBox Manager

FileMachineViewInputDevicesHelp

Ubuntu22.04.5 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

VirtualBoxORACLE

Press F12 to select boot device.

AA

Caps Lock Off

Details

Preview

General

System

Display

Storage

Audio

Network

USB

Shared folders

Description

Namesystem

Operating System

Base Memory

Processors

Boot Order

Acceleration

Video Memory

Graphics Controller

3D Acceleration

Remote Desktop Server

Recording

Controller: IDE

IDE Primary Device 0

Controller: SATA

SATA Port 0

Host Driver: Default

Controller: ICH AC97

Adapter 1: Intel PRO/1000 MT Desktop (NAT)

USB Controller: OHCI (EHCI)

Device Filters: 0 (0 active)

None

None

59°F

Mostly cloudy

Search

18:43

15/2/2026

1. Introduction

This report presents an incident investigation using the BOTSv3 dataset[1], focusing on AWS and S3 bucket events in the Frothly environment. Using Splunk as the primary analysis platform, the investigation reconstructs the cyber kill chain following the NIST Incident Response Framework [2], explains how the incident unfolded, and identifies key vulnerabilities and misconfigurations.

The primary goal of this investigation is to locate potential data leakage points in Frothly's AWS environment and understand how they could be exploited, and to derive actionable recommendations that reduce the likelihood of similar incidents in the future.

Splunk Enterprise[8] was deployed on an Ubuntu virtual machine as an independent SOC environment for ingesting and analysing BOTSv3 logs. The following sections discuss SOC roles and incident handling, installation and data preparation, guided question analysis, and finally the overall conclusions and presentation strategy.

2. SOC Roles & Incident Handling Reflection

In this report, the analysis is carried out from the perspective of a Tier 2 SOC analyst[6]handling an escalated incident.

Preventive Controls Analysis

In this incident, several preventive controls should have been in place, including mandatory MFA, least-privilege IAM policies, S3 public access blocks, and alerting on high-risk configuration changes. Without these measures, the misconfiguration of a public S3 bucket and weak authentication could easily remain unnoticed, leaving Tier 1 with no meaningful alerts to act upon, which significantly hampers the effectiveness of the initial triage phase [9].

Detection & Response

With proper prevention and monitoring, Tier 1 and Tier 2 could use CloudTrail, S3 access logs, and winhostmon data to quickly detect that a bucket had been made public and that an endpoint was behaving abnormally.

- **Tier 2 Responsibility:** Once the incident is confirmed, Tier 2 is responsible for response actions such as isolating affected resources, revoking or rolling back risky access controls, notifying the relevant owners, and documenting findings and actions.

- **Tier 3 Responsibility:** Tier 3 then uses this report as input for the recovery and improvement phase, ensuring that new access controls are fully implemented and that lessons learned are translated into updated security policies, detection rules, and playbooks.

Conclusion on SOC Visibility

Currently, BOTSv3 provides raw CloudTrail and S3 access logs, it seems without predefined alerts. In a real SOC, these patterns would be implemented as alerts.

3. Installation & Data Preparation

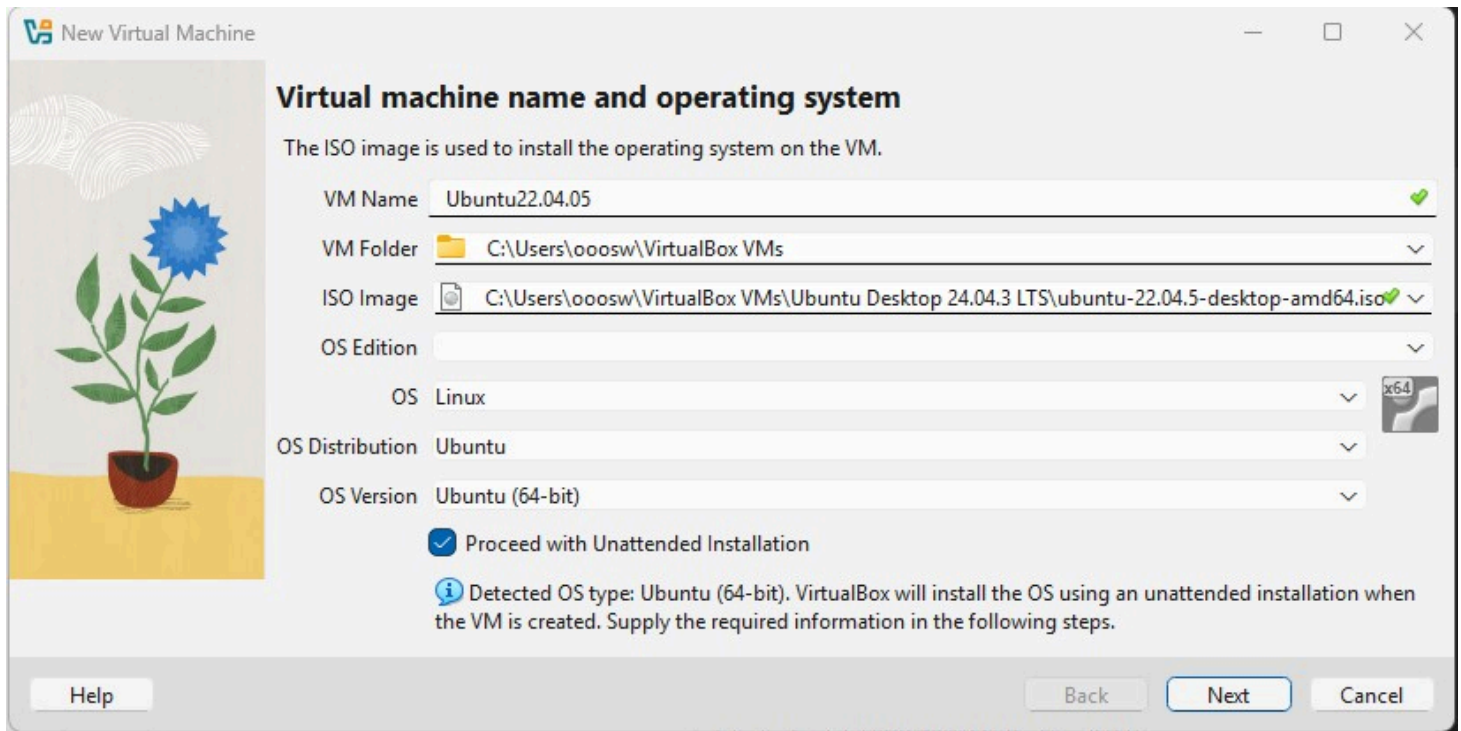
For the SOC infrastructure, an Ubuntu 22.04.5 virtual machine was deployed on a Windows host, and Splunk Enterprise 10.0.2 was installed on this VM for the investigation. This setup emulates a typical SOC architecture where a dedicated Linux server hosts the SIEM platform.

Using a VM-based Splunk deployment provides a controllable and isolated environment, separating security log analysis from the local operating system. This aligns with SOC best practices, where critical monitoring components are hosted on dedicated infrastructure rather than analyst workstations. In this lab, Splunk acts as the central SIEM, responsible for receiving, indexing, and querying security events, effectively covering both the data receiver and data enquiry roles.

The following sections describe the VM installation steps and data preparation process in detail.

- **OS:** Ubuntu 22.04.5
- **SIEM:** Splunk Enterprise 10.0.2
- **Data:** BOTSv3 Dataset

Installation Evidence



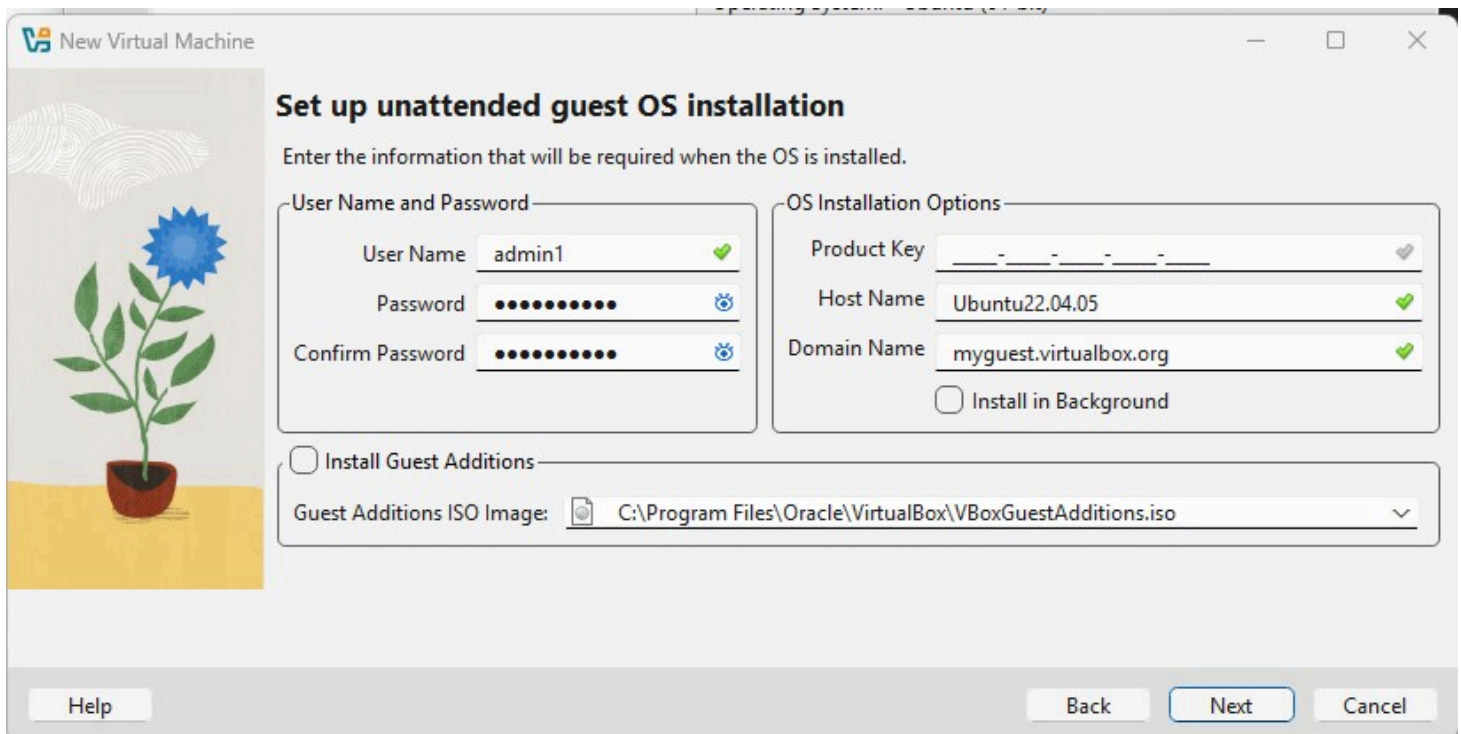
The screenshot shows the 'New Virtual Machine' wizard in Oracle VM VirtualBox. The title bar says 'New Virtual Machine'. The main heading is 'Virtual machine name and operating system'. Below it, a subtitle reads: 'The ISO image is used to install the operating system on the VM.' The form contains the following fields and values:

- VM Name: Ubuntu22.04.05 (with a green checkmark icon)
- VM Folder: C:\Users\ooosw\VirtualBox VMs (with a folder icon)
- ISO Image: C:\Users\ooosw\VirtualBox VMs\Ubuntu Desktop 24.04.3 LTS\ubuntu-22.04.5-desktop-amd64.iso (with a green checkmark icon)
- OS Edition: (empty dropdown)
- OS: Linux (with a dropdown arrow)
- OS Distribution: Ubuntu (with a dropdown arrow)
- OS Version: Ubuntu (64-bit) (with a dropdown arrow)
- Proceed with Unattended Installation: ☒ (with a blue checkmark icon)

Below the 'Proceed with Unattended Installation' checkbox, there is an information icon (i) and a message: 'Detected OS type: Ubuntu (64-bit). VirtualBox will install the OS using an unattended installation when the VM is created. Supply the required information in the following steps.'

At the bottom, there are three buttons: 'Help', 'Back', and 'Next' (highlighted with a blue border), and 'Cancel'.

Figure 3.1: Configure the VM name and select the Ubuntu ISO image



The screenshot shows the 'New Virtual Machine' wizard in Oracle VM VirtualBox, specifically the 'Set up unattended guest OS installation' step. The title bar says 'New Virtual Machine'. The main heading is 'Set up unattended guest OS installation'. Below it, a subtitle reads: 'Enter the information that will be required when the OS is installed.' The form is divided into two main sections:

- User Name and Password:**
 - User Name: admin1 (with a green checkmark icon)
 - Password: (masked with dots, with a blue eye icon to toggle visibility)
 - Confirm Password: (masked with dots, with a blue eye icon to toggle visibility)
- OS Installation Options:**
 - Product Key: (empty field with a grey heart icon)
 - Host Name: Ubuntu22.04.05 (with a green checkmark icon)
 - Domain Name: myguest.virtualbox.org (with a green checkmark icon)
 - Install in Background: ☐ (checkbox icon)

Below these sections, there is a checkbox for 'Install Guest Additions' which is currently unchecked. Below that is a field for 'Guest Additions ISO Image' with the value: C:\Program Files\Oracle\VirtualBox\VBxGuestAdditions.iso (with a dropdown arrow).

At the bottom, there are three buttons: 'Help', 'Back', and 'Next' (highlighted with a blue border), and 'Cancel'.

Figure 3.2: Configure the VM username and password

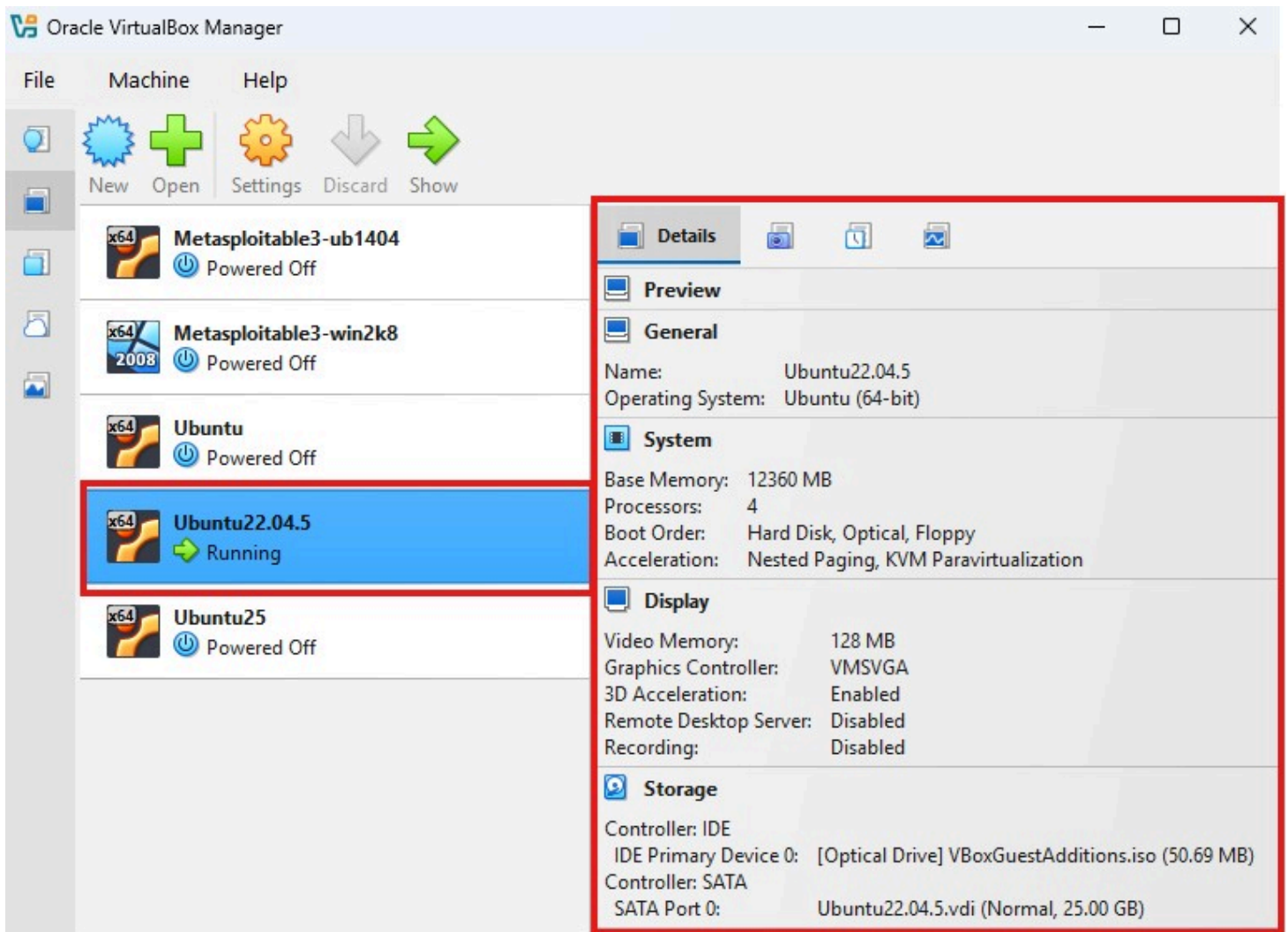


Figure 3.3: Allocate sufficient processors, RAM, and disk space to the VM

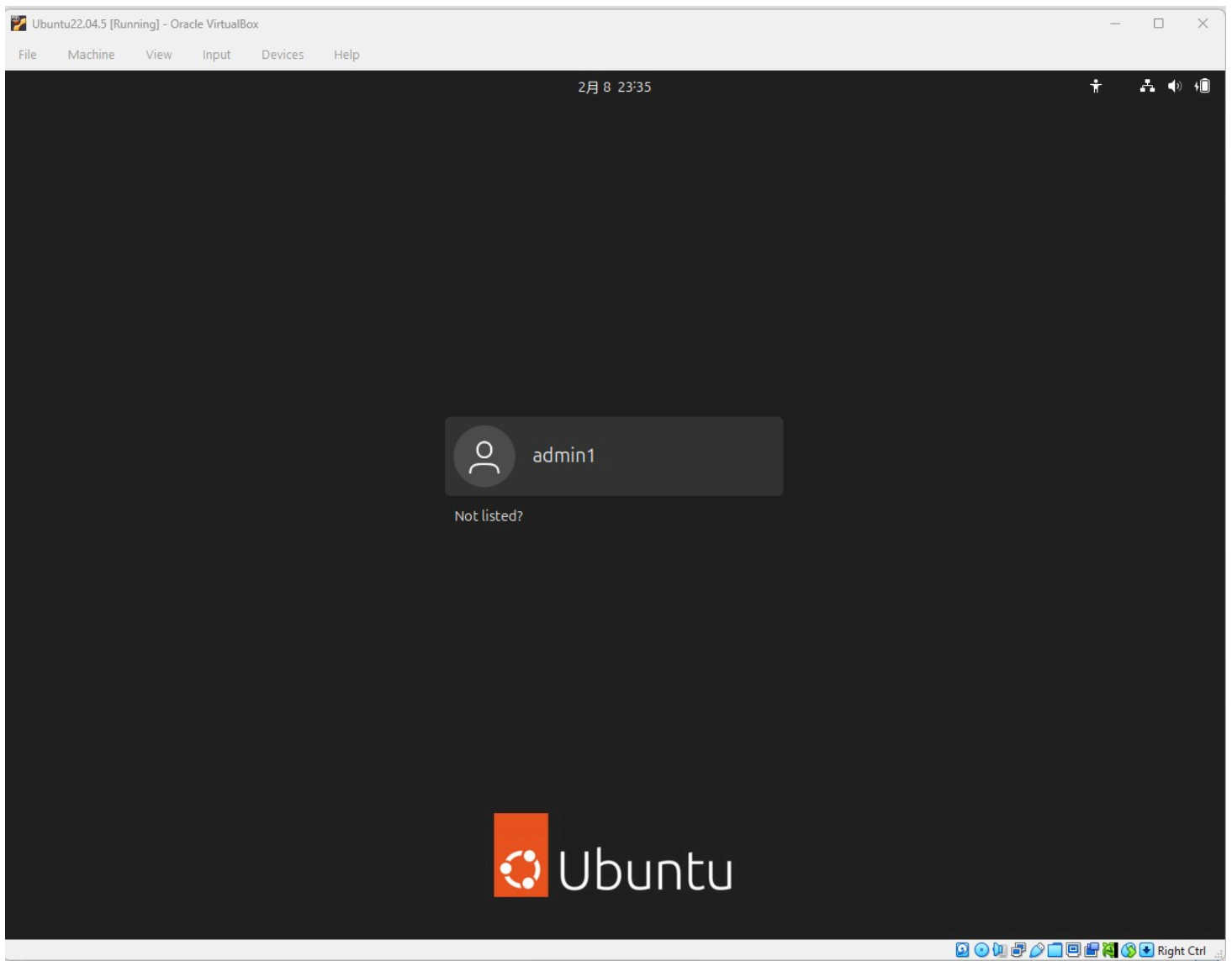


Figure 3.4: Login to the Ubuntu VM

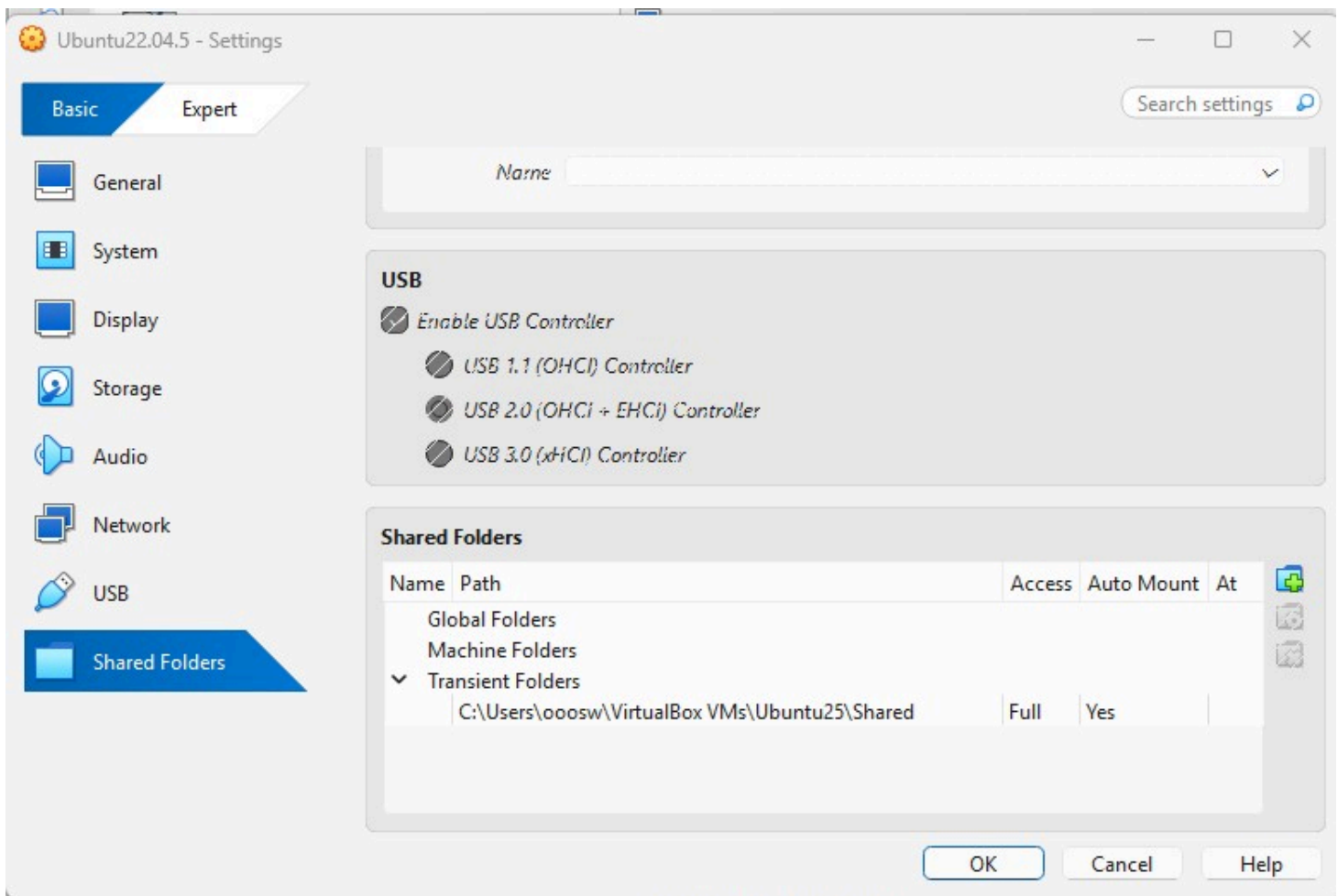


Figure 3.5: Configure VirtualBox shared folders for file transfer

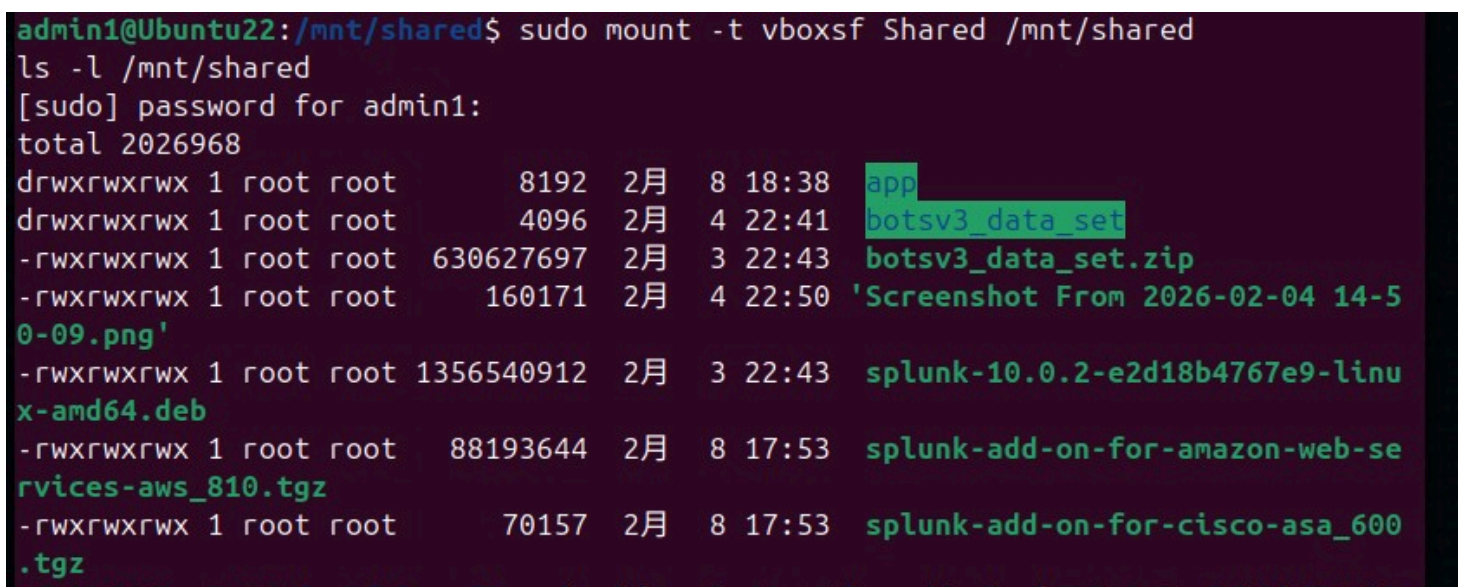


Figure 3.6: Mount the shared folder in the VM and verify its contents

```

admin1@Ubuntu22:/mnt/shared$ sudo dpkg -i splunk-10.0.2-e2d18b4767e9-linux-amd64
.deb
Selecting previously unselected package splunk.
(Reading database ... 297126 files and directories currently installed.)
Preparing to unpack splunk-10.0.2-e2d18b4767e9-linux-amd64.deb ...
verify that this system has all the commands we will require to perform the preflight
step
no need to run the splunk-preinstall upgrade check
Unpacking splunk (10.0.2) ...
Setting up splunk (10.0.2) ...
find: '/opt/splunk/lib/python3.7/site-packages': No such file or directory
complete
admin1@Ubuntu22:/mnt/shared$ ls /opt
ls /opt/splunk
splunk  VBoxGuestAdditions-7.2.4
bin      openssl
copyright.txt  opt
etc      quarantined_files
ftr      README-splunk.txt
include  share
lib      splunk-10.0.2-e2d18b4767e9-linux-amd64-manifest
license-eula.txt  swidtag
LICENSE.txt
admin1@Ubuntu22:/mnt/shared$

```

Figure 3.7: Verify that the Splunk package is suitable for installation using dpkg

```

admin1@Ubuntu22:/mnt/shared$ sudo /opt/splunk/bin/splunk start --accept-license

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise,
you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: admin
Password must contain at least:
    * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openlda
p/ldap.conf'.
writing RSA key

```

Figure 3.8: Install Splunk Enterprise and create the administrator profile

```
Warning: ignoring -extensions option without -extfile
Certificate request self-signature ok
subject=CN = Ubuntu22, O = SplunkUser
Done

Waiting for web server at http://127.0.0.1:8000 to be available..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://Ubuntu22:8000

admin1@Ubuntu22:/mnt/shared$
```

Figure 3.9: Start the Splunk service

```
admin1@Ubuntu22:/opt/splunk/bin$ sudo tar zxvf /mnt/shared/botsv3_data_set.tgz -C /opt/splunk/etc/apps
ls /opt/splunk/etc/apps
botsv3_data_set/
botsv3_data_set/lookups/
botsv3_data_set/LICENSE
botsv3_data_set/bin/
botsv3_data_set/var/
botsv3_data_set/default/
botsv3_data_set/README.txt
botsv3_data_set/default/app.conf
botsv3_data_set/default/indexes.conf
botsv3_data_set/default/tags.conf
botsv3_data_set/default/props.conf
```

Figure 3.10: Import the BOTSV3 dataset into Splunk

```

botsv3_data_set/lookups/scanner_agents.csv
botsv3_data_set/lookups/ransomware_extensions.csv
botsv3_data_set/lookups/ddns_provider.csv
botsv3_data_set/lookups/iis_action_lookup.csv
alert_logevent          splunk-data-management
alert_webhook           SplunkDeploymentServerConfig
appsbrowser            SplunkForwarder
audit_trail             splunk_gdi
botsv3_data_set         splunk_httpinput
introspection_generator_addon splunk_instrumentation
journald_input          splunk_internal_metrics
launcher               SplunkLightForwarder
learned                 splunk_metrics_workspace
legacy                 splunk_monitoring_console
python_upgrade_readiness_app splunk_pipeline_builders
sample_app             splunk_rapid_diag
search                  splunk-rolling-upgrade
splunk_app_for_splunk_o11y_cloud splunk_secure_gateway
splunk_archiver         splunk-visual-exporter
splunk-dashboard-studio user-prefs

```

Figure 3.11: Verify that the BOTSv3 dataset was successfully ingested

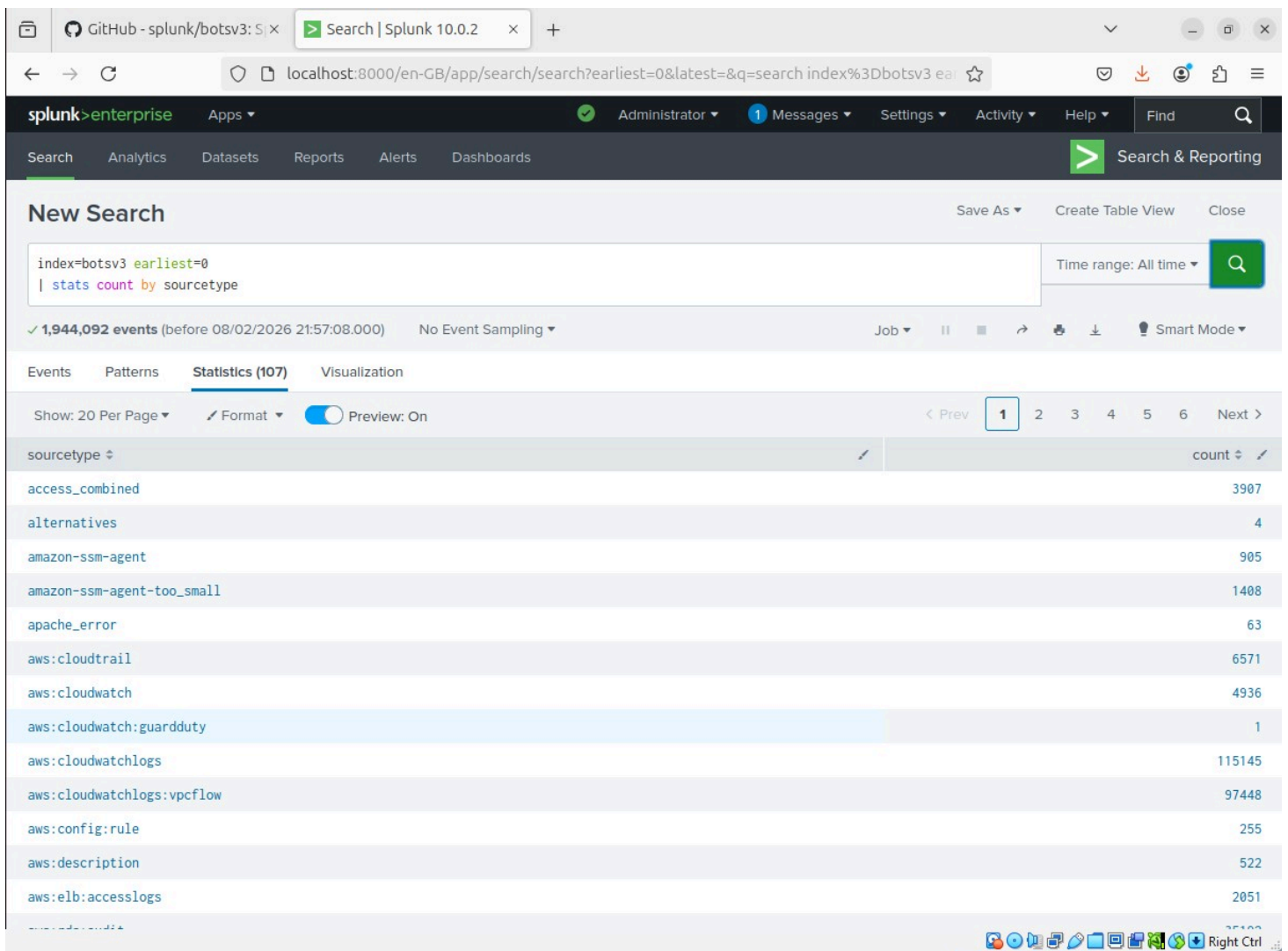


Figure 3.12: Log in to Splunk and run an initial search on the BOTSV3 index

Data Validation

In this environment, the validation query `index=botsv3 earliest=0 | stats count by sourcetype` shows that the botsv3 index contains 107 sourcetypes, including critical ones such as `aws:cloudtrail` and `winhostmon`. This confirms that both key cloud audit logs and host telemetry are available for the subsequent SOC investigation. From a SOC perspective, validating sourcetype coverage, event volumes, and field extraction is a critical onboarding step before building detections and dashboards on top of a new data source.

While this all-in-one Splunk deployment on a single Ubuntu VM is sufficient for this incident analysis, it has limitations compared to a production environment. It does not separate indexer and search head roles. These are acceptable in a lab context because they preserve the core SIEM concepts—centralised logging, correlation, and tiered analysis—while keeping the environment simple enough to deploy and replicate for this report.

4. Guided Questions Analysis

In a production environment, incident analysis typically begins with an automated alert. This report adopts the persona of a Tier 2 SOC Analyst at **Frothly** who has received a ticket regarding suspicious configuration changes within the organization's Amazon AWS cloud infrastructure.

The objective of this section is to conduct a forensic investigation using Splunk to determine the full scope of the incident. This includes identifying the trigger point, tracking the suspect via Identity & Access Management (IAM), assessing the scope of impact which is public data exposure, and capturing infrastructure evidence to support remediation.

Question 1: Identifying IAM Users

The investigation began by identify who exists in the environment queried CloudTrail logs to list all unique user identities to understand who exists in the environment.

Splunk Query:

```
index=botsv3 sourcetype="aws:cloudtrail" userIdentity.type="IAMUser"
| stats count by userIdentity.userName
```

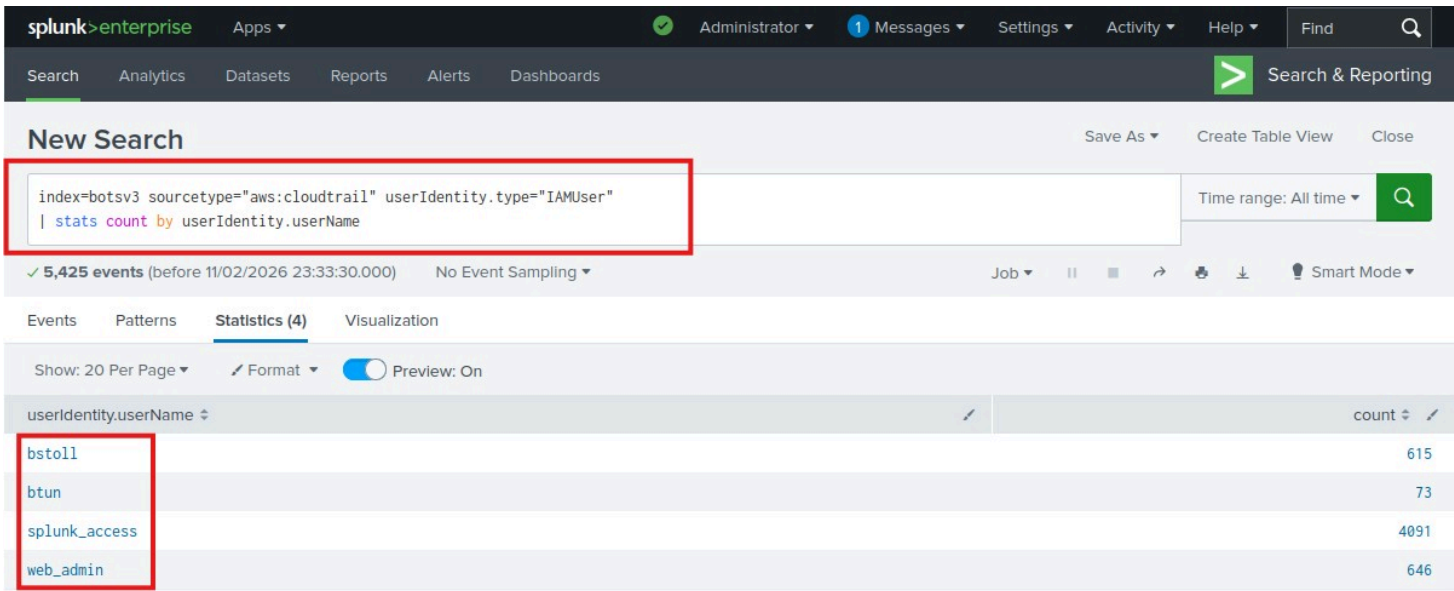


Figure 4.1

The IAM users identified are:\n bstoll, btun, splunk_access, web_admin

In SOC environment, any account found here that does not map to a current list would be flagged for immediate disablement.

Question 2: Multi-Factor Authentication Check

After identify who exists in the environment drilled down into the JSON of the CloudTrail logs to check the status of Multi-Factor Authentication (MFA) Status.

The specific JSON path used to identify non-MFA sessions is:

userIdentity.sessionContext.attributes.mfaAuthenticated.

The screenshot shows the Splunk search interface. The search bar contains the query: `index=botstv3 sourcetype="aws:cloudtrail"`. The results are displayed in a table with columns: **i**, **Time**, and **Event**. The first result is expanded, showing the JSON data for a CloudTrail event. The event details are as follows:

Field	Value
awsRegion	us-west-1
eventID	51b2664e-dd61-4db3-ab77-31b649e9c69d
eventName	DescribeSecurityGroups
eventSource	ec2.amazonaws.com
eventTime	2018-08-20T15:15:04Z
eventType	AwsApiCall
eventVersion	1.05
recipientAccountId	622676721278
requestID	11a2f0a0-78b1-4968-8887-43ab8a9ba0d5
requestParameters	{ }
responseElements	null
sourceIPAddress	107.77.212.175
userAgent	signin.amazonaws.com
userIdentity	{ accessKeyId: ASIAZB6TMXZ7FYCAEHNH accountId: 622676721278 arn: arn:aws:iam::622676721278:user/bstoll invokedBy: signin.amazonaws.com principalId: AIDAJUFKXZ44LV4EN4MGK sessionContext: { attributes: { creationDate: 2018-08-20T15:04:44Z mfaAuthenticated: false } } type: IAMUser userName: bstoll }

Red arrows point to the `userIdentity`, `sessionContext`, and `mfaAuthenticated: false` fields in the JSON output.

Figure 4.2.1

Splunk Query:

`index=botstv3 sourcetype="aws:cloudtrail"`

```
|spath output=mfa_status path=userIdentity.sessionContext.attributes.mfaAuthenticated  
|search mfa_status=false  
|stats count by userIdentity.userName  
|sort - count
```

The Figure 4.2.1 is showing the MFA status as false, and though the query Figure 4.2.2 show the all the user who the MFA status as false and count by descending.

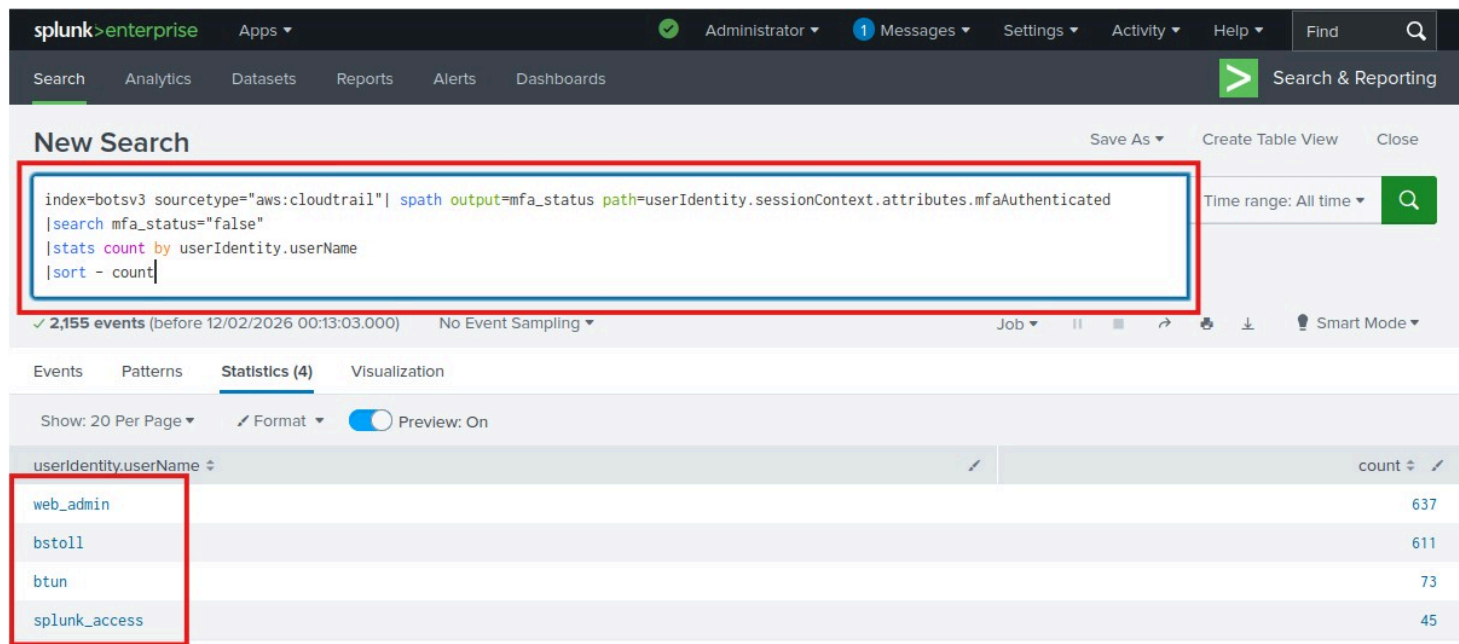


Figure 4.2.2

In a production SOC, lack of MFA on administrative accounts is a critical vulnerability, and CISA [7] explicitly recommends phishing-resistant MFA for all privileged cloud accounts."

Question 3: Hardware Inventory

After identifying who can access this environment, the second step is to identify the web servers' hardware specifications for asset inventory. This pivots to WinHostMon and hardware data sources, enabling correlation between incidents and physical assets.

Splunk Query:

```
index=botsv3 sourcetype="WinHostMon"  
|stats count by Type
```

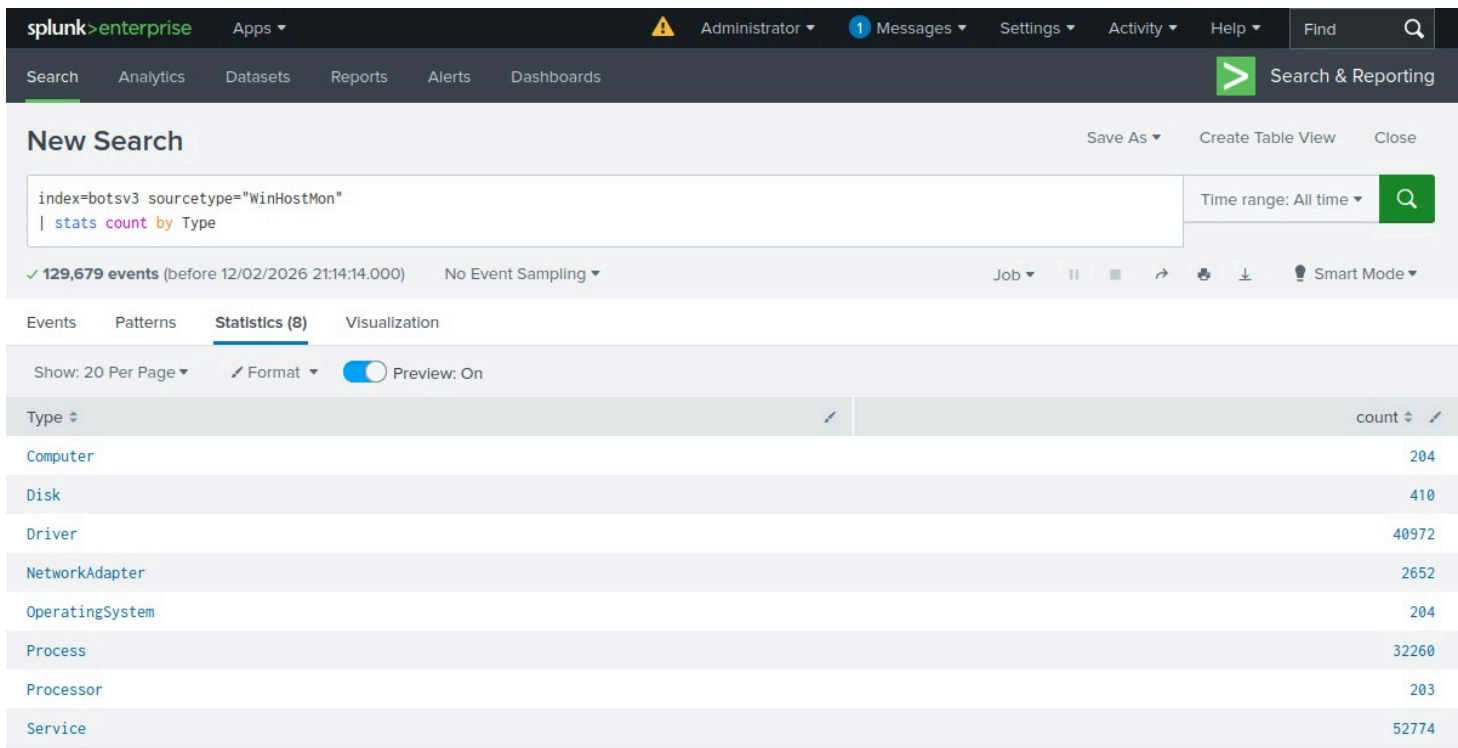


Figure 4.3.1

Figure 4.3.1 shows the WinHostMon event counts by Type (processor, disk, logical), representing Windows host monitoring information from various terminals.

Splunk Query:

```
index=botsv3 sourcetype=hardware
| stats count by host
```

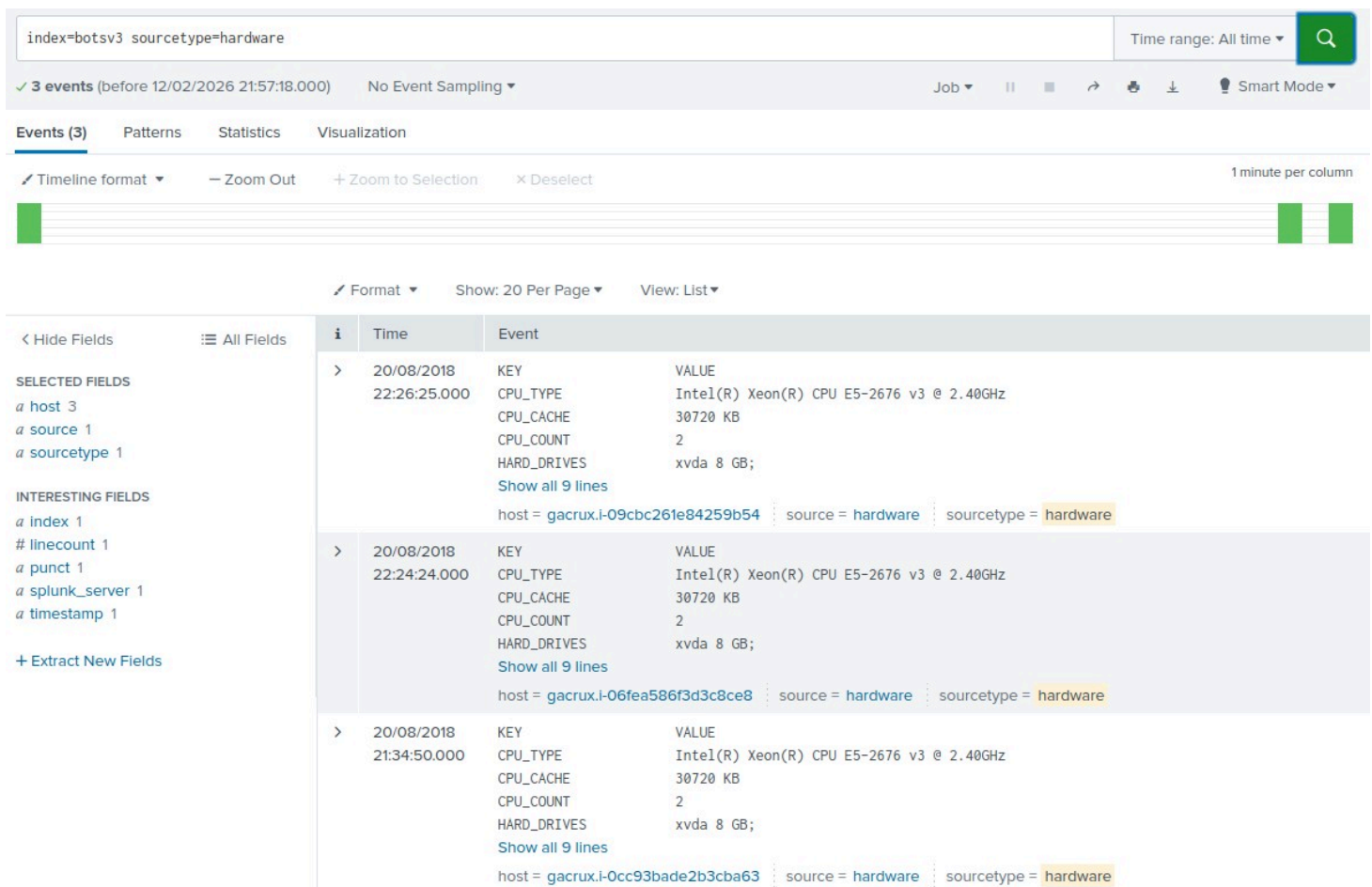


Figure 4.3.2

Figure 4.3.2 lists all web servers in the dataset, all using Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz.

Hardware inventory is critical in a SOC and in digital forensics because it provides a complete catalog of system assets. It helps organizations identify the scope of affected devices and assess their risk exposure. In digital forensics, it can confirm the system configuration at the time of an incident and support device authenticity verification.

Question 4-6: S3 bucket

It is known that this incident was triggered by an S3 bucket being configured as public. To investigate this, I utilized the aws:cloudtrail sourcetype, which is the standard log source for recording AWS API calls [5]. By searching these logs, I located the specific Event ID to determine the trigger time, identify the user who performed the misconfiguration, and confirm the S3 bucket name.

Splunk Query:

```
index=botsv3 sourcetype="aws:cloudtrail" eventName="PutBucketAcl" "AllUsers"
| table eventTime eventSource eventName requestParameters eventide
```

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

index=botsv3 sourcetype=aws:cloudtrail" eventName="PutBucketAc1" "AllUsers"
 | table eventTime eventSource eventName requestParameters eventID

Time range: All time

✓ 1 event (before 12/02/2026 22:41:29.000) No Event Sampling Job Visualization

Events Patterns Statistics (1) Visualization

Show: 20 Per Page Format Preview: On

eventTime	eventSource	eventName	requestParameters	eventID
2018-08-20T13:01:46Z	s3.amazonaws.com	PutBucketAc1		ab45689d-69cd-41e7-8705-5350402cf7ac

Figure 4.4.1

```

    {
      [-]
      Grantee: { [-]
        URI: http://acs.amazonaws.com/groups/global/AllUsers
        xmlns:xsi: http://www.w3.org/2001/XMLSchema-instance
        xsi:type: Group
      }
      Permission: READ
    }
    {
      [-]
      Grantee: { [-]
        URI: http://acs.amazonaws.com/groups/global/AllUsers
        xmlns:xsi: http://www.w3.org/2001/XMLSchema-instance
        xsi:type: Group
      }
      Permission: WRITE
    }
  ]
}
Owner: { [-]
  DisplayName: bstoll
  ID: 4c018053e740f45beb45f68c0f5eff6347745488ae540130432c9fc64fae310d
}
xmlns: http://s3.amazonaws.com/doc/2006-03-01/
}
acl: [ [+]
]
bucketName: frothlywebcode
}
responseElements: null
sourceIPAddress: 107.77.212.175
userAgent: signin.amazonaws.com

```

Figure 4.5.1

```

    }
    Owner: { [-]
      DisplayName: bstoll
      ID: 4c018053e740f45beb45f68c0f5eff6347745488ae540130432c9fc64fae310d
    }
    xmlns: http://s3.amazonaws.com/doc/2006-03-01/
  }
  acl: [ [+]
  ]
  bucketName: frothlywebcode
}
responseElements: null
sourceIPAddress: 107.77.212.175
userAgent: signin.amazonaws.com
userIdentity: { [-]
  accessKeyId: ASIAZB6TMXZ70A2RDK5X
  accountId: 622676721278
  arn: arn:aws:iam::622676721278:user/bstoll
  invokedBy: signin.amazonaws.com
  principalId: AIDAJUFKXZ44LV4EN4MGK
  sessionContext: { [-]
    attributes: { [-]
      creationDate: 2018-08-20T12:19:44Z
      mfaAuthenticated: false
    }
  }
}
type: IAMUser
userName: bstoll

```

Figure 4.6.1

In Figure 4.4.1, the query lists all eventName="PutBucketAcl" events where the grantee is AllUsers within sourcetype="aws:cloudtrail". In this incident, the relevant eventID is **"ab45689d-69cd-41e7-8705-5350402cf7ac"**

When drilling down into the JSON event under requestParameters, there are two grantees. For each grantee, the uri is set to AllUsers, and the permissions are set to READ and WRITE (with xsi:type set to Group). This indicates that the event configured the S3 bucket ACL to be publicly accessible, violating AWS Security Best Practices [3].

In the same JSON event shown in Figures 4.5.1 and 4.6.1, the configuration applies to the bucketName: **"frothlywebcode"** and the userName: **"bstoll"**. This indicates that on **"2018-08-20 at 13:01:46"**, user bstoll configured the frothlywebcode bucket, resulting in it being accessible by everyone.

Question 7: Impact Assessment

After identifying that the "frothlywebcode" bucket has been misconfigured, as a SOC Tier 2 analyst, I am required to determine if the exposure was exploited. Use the following query to look for PUT and GET requests from external IP addresses targeting the "frothlywebcode" bucket.

Splunk Query:

```
index=botsv3 sourcetype="aws:s3:accesslogs" "frothlywebcode" "PUT"
| rex field=_raw "(?<client_ip>\d+\.\d+\.\d+\.\d+)"
| rex field=_raw "^S+\s+(?S+)\s+["
| rex field=_raw "\"PUT\s+/(?<object_key>[^\s]+\s+HTTP/1.1\""
| table _time client_ip bucket object_key
```

```

**Splunk Query:**\
index=botsv3 sourcetype="aws:s3:accesslogs" "frothlywebcode" "PUT"
| rex field=_raw "(?<client_ip>\d+\.\d+\.\d+\.\d+)"
| rex field=_raw "^S\s+(?<bucket>\S+)\s+\["
| rex field=_raw "\"PUT\s+/(?<object_key>[^ ]+)\s+HTTP/1\.\d\""
| table _time client_ip bucket object_key

```

\

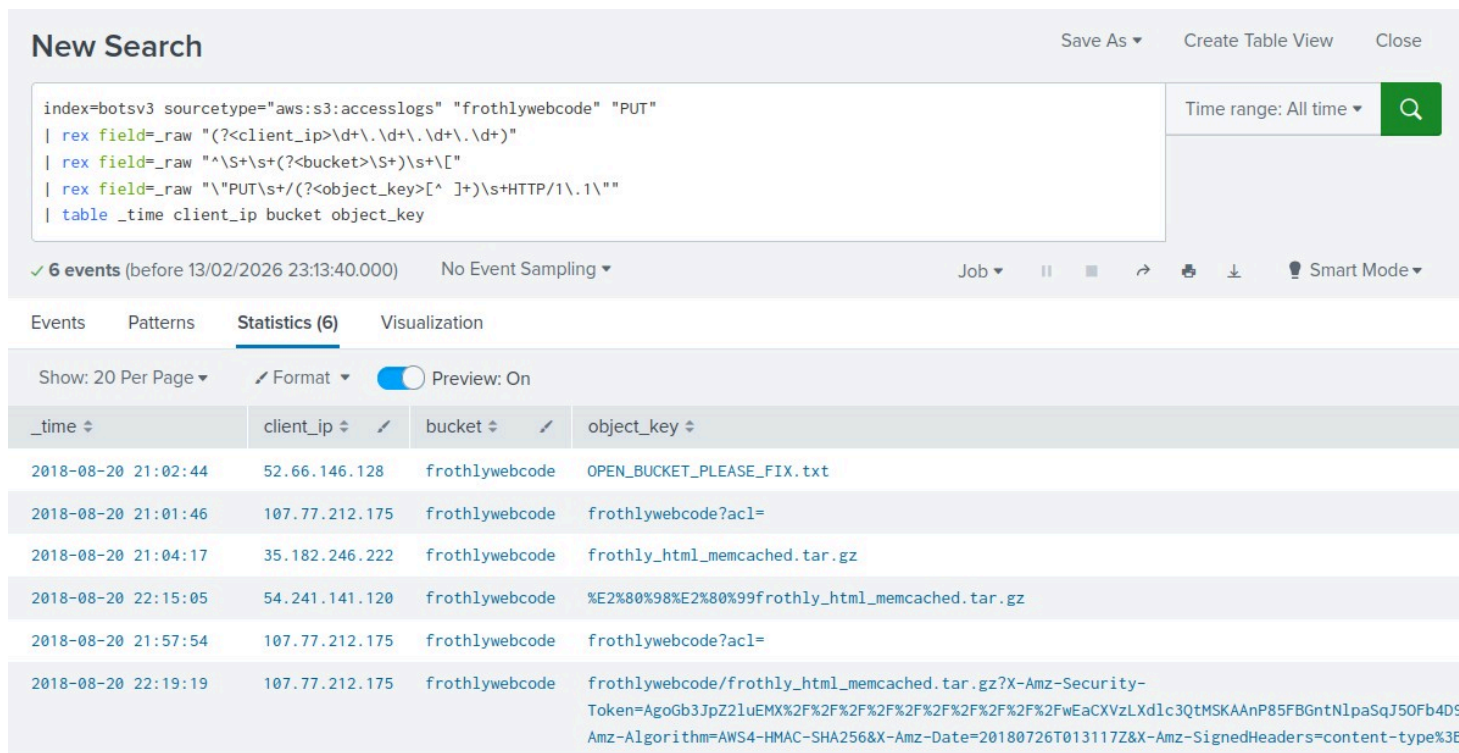


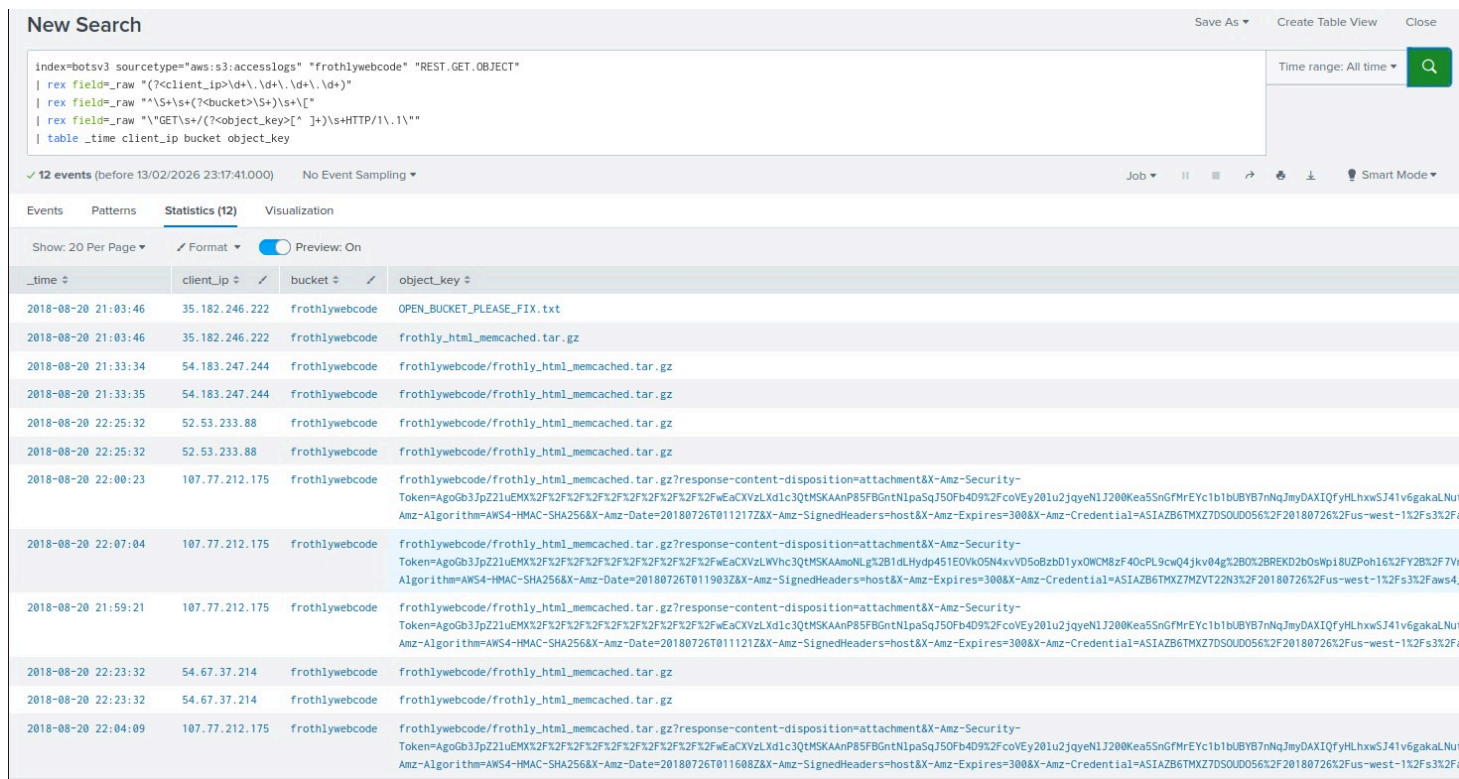
Figure 4.7.1

```
index=botsv3 sourcetype="aws:s3:accesslogs" "frothlywebcode" "REST.GET.OBJECT"
| rex field=_raw "(?<client_ip>\d+\.\d+\.\d+\.\d+)"
| rex field=_raw "^S+\s+(?S+)\s+["
| rex field=_raw "\"GET\s+/(?<object_key>[^ ]+)\s+HTTP/1.1\""
| table _time client_ip bucket object_key
```

```

**Splunk Query:**\
index=botsv3 sourcetype="aws:s3:accesslogs" "frothlywebcode" "REST.GET.OBJECT"
| rex field=_raw "(?<client_ip>\d+.\d+.\d+.\d+)"
| rex field=_raw "^S\s+(?<bucket>S+)\s+\["
| rex field=_raw "\"GET\s+/(?<object_key>[^ ]+)\s+HTTP/1\.\d\""
| table _time client_ip bucket object_key

```



In Figure 4.7.1 is show someone put a. text file **“OPEN_BUCKET_PLEASE_FIX.txt”** into the bucket from IP address 52.66.146.128, and IP address 35.182.46.222 put the **“frothly_html_memcached.tar.gz”** into the bucket. From Figure 4.7.3 in the event detail of IP address 107.77.212.175 can prove the user as bstoll configure the acl and using the pre-signed URL to upload the **“frothly_html_memcached.tar.gz”** From Figure 4.7.4.

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

```
index=botsv3 sourcetype="WinHostMon"
| dedup host
| table host
| sort host
```

✓ 8 events (before 14/02/2026 02:06:40.000) No Event Sampling ▾

Events Patterns **Statistics (8)** Visualization

Show: 20 Per Page ▾ / Format ▾ ☒ Preview: On

host ⇅

ABUNGST-L
BGIST-L
BSTOLL-L
BTUN-L
FYODOR-L
JWORTOS-L
MKRAEUS-L
PCERF-L

Figure 4.8.1

Splunk Query:

index=botsv3 "BSTOLL-L" host="BSTOLL-L"

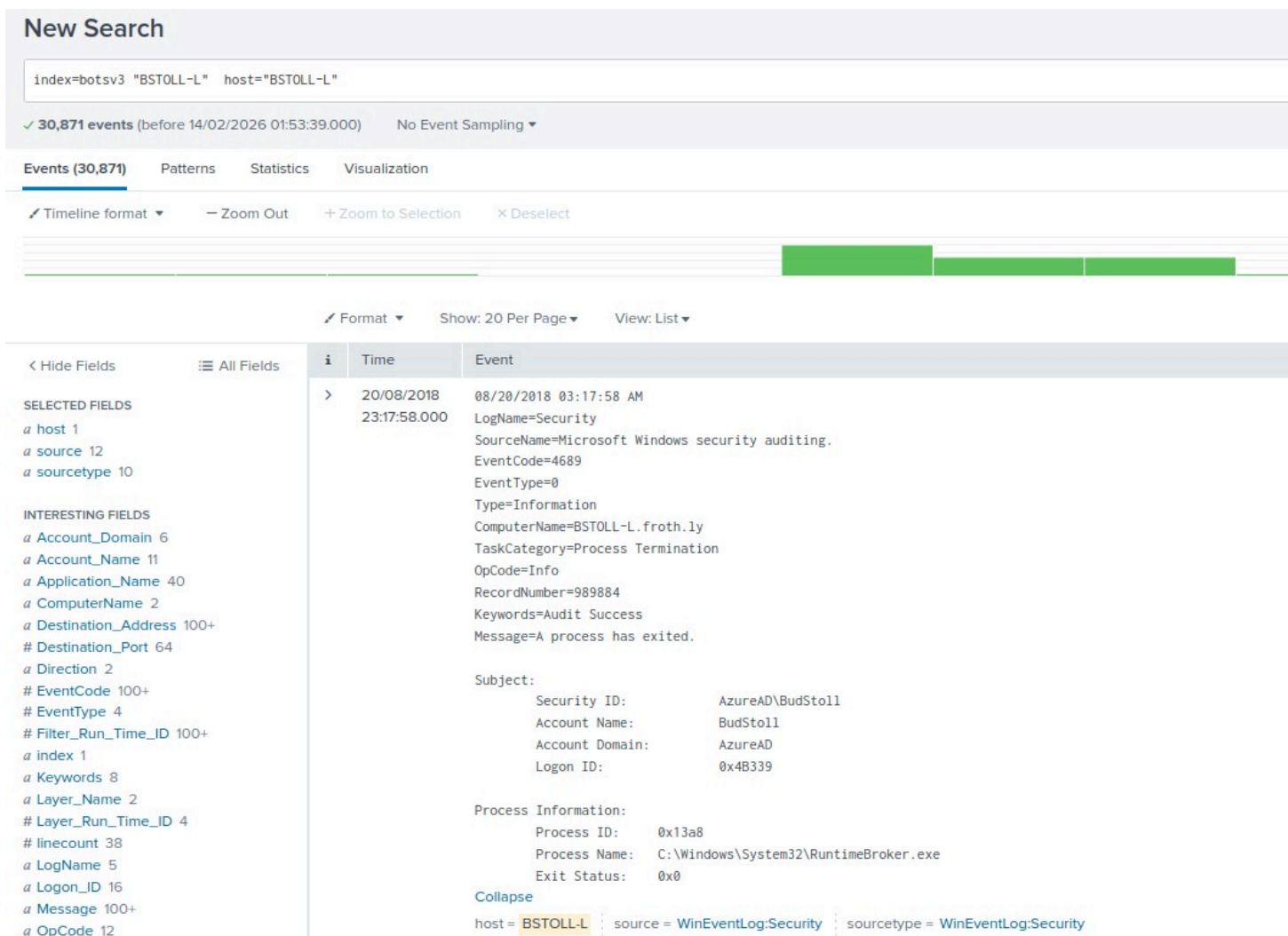


Figure 4.8.2

Figure 4.8.1 shows all hosts that have been identified, and Figure 4.8.2 shows that the FQDN of host "BSTOLL-L" is "BSTOLL-L.froth.ly"

Splunk Query:

```
index=botsv3 sourcetype="WinHostMon" Type=OperatingSystem
| stats count by host, OS
```

New Search			Save As ▾	Create Table View	Close
index=botsv3 sourcetype="WinHostMon" Type=OperatingSystem stats count by host, OS			Time range: All time ▾ 		
✓ 204 events (before 14/02/2026 02:21:22.000) No Event Sampling ▾			Job ▾ ↗ ⚙ ⬇ ⚡ Smart Mode ▾		
Events Patterns Statistics (8) Visualization					
Show: 20 Per Page ▾ ✓ Format ▾ ☒ Preview: On					
host ↕	OS ↕	count ↕			
BSTOLL-L	Microsoft Windows 10 Enterprise	30			
ABUNGST-L	Microsoft Windows 10 Pro	16			
BGIST-L	Microsoft Windows 10 Pro	23			
BTUN-L	Microsoft Windows 10 Pro	29			
FYODOR-L	Microsoft Windows 10 Pro	23			
JWORTOS-L	Microsoft Windows 10 Pro	30			
MKRAEUS-L	Microsoft Windows 10 Pro	25			
PCERF-L	Microsoft Windows 10 Pro	28			

Figure 4.8.3

Figure 4.8.3 shows which operating system each host uses. BSTOLL-L is using Microsoft window 10 Enterprise.

5. Conclusion, References and Presentation

This investigation confirmed a data exposure incident within Frothly's AWS environment. The root cause was identified as a misconfiguration by user bstoll, who modified the frothlywebcode S3 bucket's Access Control List (ACL) to public via the PutBucketAcl API (Event ID: ab45689d-69cd-41e7-8705-5350402cf7ac).

This exposure allowed external actors to upload a warning file, 'OPEN_BUCKET_PLEASE_FIX.txt', and a potentially malicious archive, 'frothly_html_memcached.tar.gz', which could be downloaded, modified, and re-uploaded. Analysis of the associated endpoint (BSTOLL-L) revealed indicators of cryptojacking, specifically sustained 100% CPU usage by browser processes, suggesting the user's compromise extended from the cloud environment to the physical endpoint.

The incident highlights that while logs were being collected, a lack of active alerting prevented Tier 1 analysts from responding. Furthermore, the absence of Multi-Factor Authentication (MFA) for multiple users represents a fundamental failure in identity security.

Recommendations:

- 1.Enable 'Block Public Access' at the AWS account level to prevent accidental bucket exposure.
- 2.Enforce MFA for all IAM users and rotate bstoll's credentials immediately.
- 3.Create high-severity Splunk alerts for specific CloudTrail events (like PutBucketAcl) and endpoint alerts for sustained high CPU usage associated with known mining pools to detect cryptojacking."

Reference

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- [2] P. Cichonski, T. Millar, T. Grance, and K. Scarfone, "Computer Security Incident Handling Guide," *National Institute of Standards and Technology (NIST)*, Gaithersburg, MD, Special Publication 800-61 Rev. 2, Aug. 2012. doi: 10.6028/NIST.SP.800-61r2.
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- [8] Splunk, "Splunk Enterprise Documentation: About Splunk Enterprise," *Splunk Docs*, 2024. [Online]. Available: <https://docs.splunk.com/Documentation/Splunk/latest/Overview/AboutSplunkEnterprise>.
- [9] E. D. Kilgallon, "Improving the Effectiveness of the Tier 1 Security Analyst," *SANS Reading Room*, 2020.

Appendix A - Cryptojacking

(This section provides supplementary forensic analysis beyond the guided questions)

The previous section showed that Frothly's S3 bucket was misconfigured. Since the dataset is from 2018, a period when cryptojacking was prevalent, I hypothesized that the compromised endpoint (BSTOLL-L) might be infected.

Splunk Query:

```
index=botsv3 sourcetype="PerfmonMk:Process"
| rex field=_raw "(?<extracted_process_name>[\\s]+)\\s+\\s+(?<extracted_cpu>\\s+)"
| search extracted_process_name="*"
| search extracted_cpu > 0
| stats avg(extracted_cpu) as Avg_CPU max(extracted_cpu) as Max_CPU by
extracted_process_name, host
| sort -Max_CPU
```

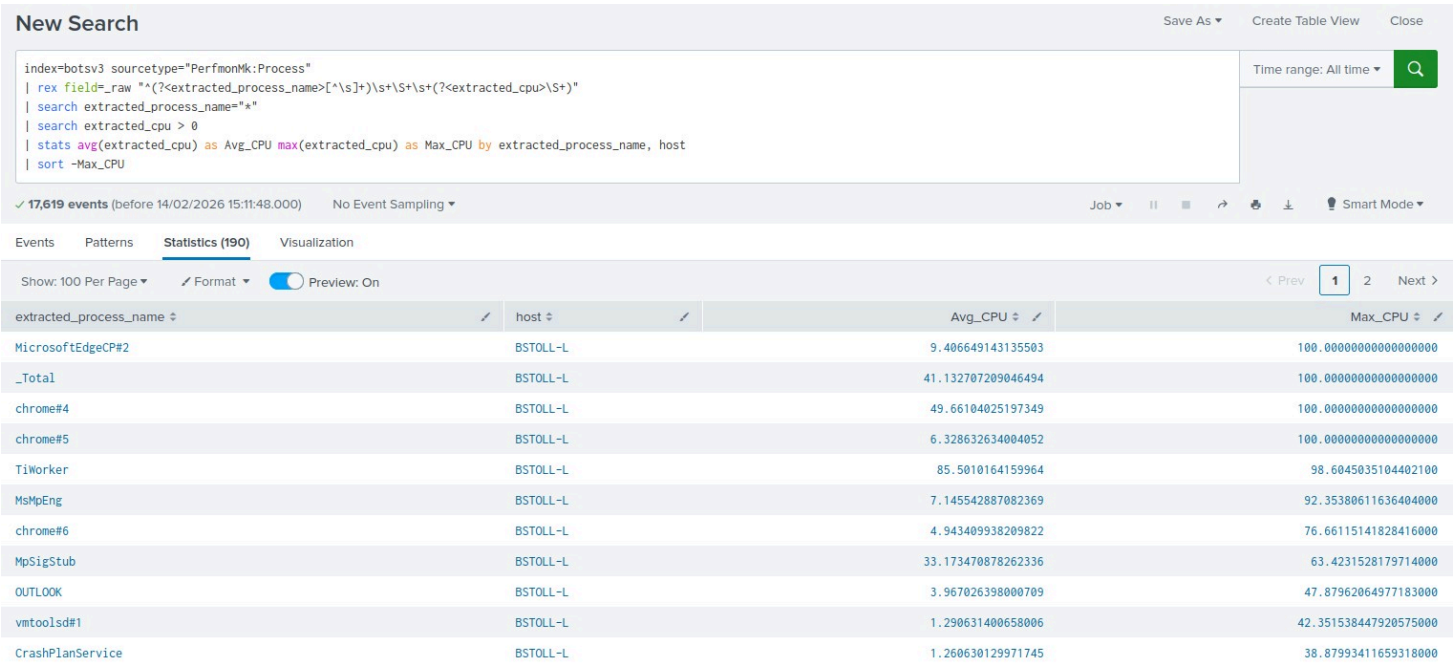


Figure 4.9.0

Figure 4.9.0 shows that the browser process is unusually consuming 100% of the CPU. Figures 4.9.1 to 4.9.3 confirm that the browser's high CPU usage time matches the DNS record, indicating a browser-based cryptojacking incident,, which aligns with MITRE ATT&CK technique T1496 (Resource Hijacking) [4].

Splunk Query:

```
index=botsv3 sourcetype="PerfmonMk:Process" host="BSTOLL-L"  
| rex field=_raw "(?<extracted_process_name>[\\s]+)\\s+\\s+(?<extracted_cpu>\\s+)"  
| search extracted_process_name IN ("_Total", "chrome#4", "chrome#5", "TiWorker", "MsMpEng",  
"chrome#6")  
| timechart span=1m max(extracted_cpu) by extracted_process_name limit=10
```

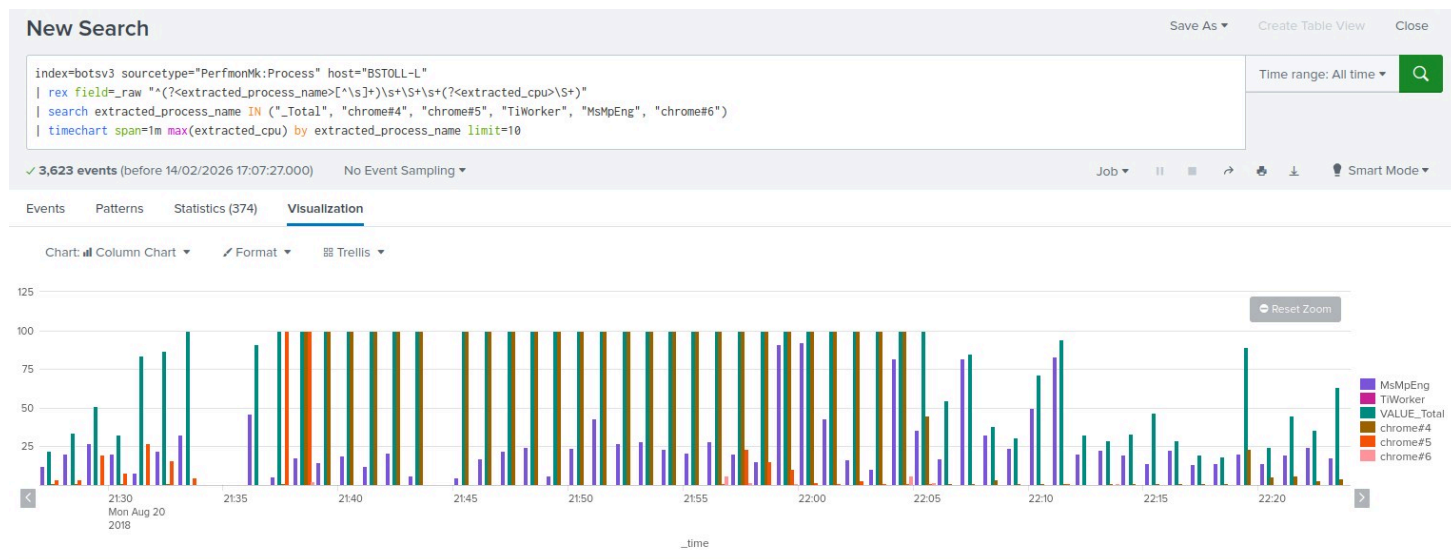


Figure 4.9.1

Splunk Query:

```
index=botsv3 sourcetype="stream:dns" query!=" " query!="-"  
earliest="08/20/2018:21:39:00"  
| table _time, query  
| sort _time
```

New Search

Save As

Create Table View

Close

index=botsv3 sourcetype="stream:dns" query!=" " query!="-"

earliest="08/20/2018:21:39:00"

| table _time, query

| sort _time

Time range: All time

5,318 events (before 14/02/2026 16:25:51.229)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (5,318)

Visualization

Show: 100 Per Page

Format

Preview: On

Prev

1

2

3

4

5

6

7

8

...

Next

_time	query
2018-08-20 21:39:02.000	splunk.froth.ly
2018-08-20 21:39:02.000	polaris.cr7wvryvz4o.us-west-1.rds.amazonaws.com
2018-08-20 21:39:02.000	splunk.froth.ly
2018-08-20 21:39:02.000	polaris.cr7wvryvz4o.us-west-1.rds.amazonaws.com
2018-08-20 21:39:02.000	gacrux.i-0cc93bade2b3cba63.us-west-1.compute.internal
2018-08-20 21:39:02.000	gacrux.i-0cc93bade2b3cba63
2018-08-20 21:39:07.000	splunk.froth.ly
2018-08-20 21:39:07.000	splunk.froth.ly
2018-08-20 21:39:10.152	splunk.froth.ly
2018-08-20 21:39:10.152	outlook.office365.com
2018-08-20 21:39:10.152	odc.officeapps.live.com
2018-08-20 21:39:10.152	nexusrules.officeapps.live.com

Figure 4.9.2

_time ↕	query ↕
2018-08-20 21:39:18.182	192.168.1.102:11-adm:arpa
2018-08-20 21:39:18.182	nexusrules.officeapps.live.com
2018-08-20 21:39:18.182	outlook.office365.com
2018-08-20 21:39:18.182	splunk.froth.ly
2018-08-20 21:39:20.185	www.mybb.com
2018-08-20 21:39:20.185	www.brewertalk.com
2018-08-20 21:39:20.185	ws019.coinhive.com
2018-08-20 21:39:20.185	wpad.froth.ly
2018-08-20 21:39:20.185	splunk.froth.ly
2018-08-20 21:39:20.185	safebrowsing.googleapis.com
2018-08-20 21:39:20.185	s3.amazonaws.com
2018-08-20 21:39:20.185	coinhive.com
2018-08-20 21:39:20.185	www.brewertalk.com
2018-08-20 21:39:20.185	s3.amazonaws.com
2018-08-20 21:39:20.185	ws019.coinhive.com
2018-08-20 21:39:20.185	splunk.froth.ly
2018-08-20 21:39:20.185	safebrowsing.googleapis.com
2018-08-20 21:39:20.185	www.mybb.com
2018-08-20 21:39:20.185	coinhive.com
2018-08-20 21:39:30.000	splunk.froth.ly
2018-08-20 21:39:30.000	mars.i-08e52f8b5a034012d.us-west-1.compute.internal
2018-08-20 21:39:30.000	mars.i-08e52f8b5a034012d
2018-08-20 21:39:30.000	splunk.froth.ly

Figure 4.9.3

Appendix B - Student Declaration of AI Tool Use

The complete Student Declaration PDF follows on the next page.

[View Student Declaration PDF](#)

Student Declaration of AI Tool use in this Assessment

Please indicate your level of usage of generative AI for this assessment - please tick the appropriate category(s).

If the “Assisted Work” or “Partnered Work” category is selected, please expand on the usage and in which elements of the assignment the usage refers to.

Solo Work	S1 - Generative AI tools have not been used for this assessment.	☐
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.	☐
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student’s own work.	☐
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student’s own work.	☐
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student’s responsibility.	☐
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.	☐
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.	☐
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.	☐
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.	☐
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge	☐

	cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.	
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.	☐
	A11 - Other uses not listed above Please specify:	☐
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify: • Drafting and Refinement • Technical Assistance & Scripting • Validation • Video Preparation	☒

Please provide details of AI usage and which elements of the coursework this relates to:

Drafting and Refinement:
 I used the AI to generate the initial structure of the report (headers and layout) and to draft specific sections, including the Introduction, SOC Roles, and Conclusion. I also used the AI to refine the grammar and professional tone of my own writing to ensure it met the "Tier 2 Analyst" persona requirements.

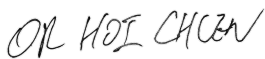
Technical Assistance & Scripting:
 The AI assisted in troubleshooting the Splunk installation on my Ubuntu VM (specifically resolving architecture issues between AMD64/ARM64). I also used the AI to generate and format the Splunk Processing Language (SPL) queries used in the Guided Questions section.

Validation
 I verified all Splunk queries by running them in my local Splunk environment to ensure they produced the correct screenshots. I also reviewed the generated report text to ensure it matched my findings.

Video Preparation:
 The AI helped outline the script and timing for the video presentation to ensure it covered all assessment criteria within the 10-minute limit.

Verification:
 I personally verified all Splunk search results by running them in my local environment and taking my own screenshots. The final analysis is my own work.

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒


 OR Hoichuen 10957621

Appendix C - GitHub Repository Availability & Incident Report

Date of Report: 16 Feb 2026

GitHub Username: 10957621

Repository URL: https://github.com/10957621/COMP3010HK_ASSESSMENT_CW1-CW2

1. Incident Summary

On 2026-02-16 12:38 UTC, I received an email from GitHub with the subject "**Password reset notification due to suspicious activity**". Therefore, I believe my GitHub account was flagged by GitHub's automated security systems due to a detected unauthorized login attempt (Account Takeover Protection). As a result, my public profile and repositories are currently hidden from public view and return a 404 error. I reset the password and enable the MFA immediately.

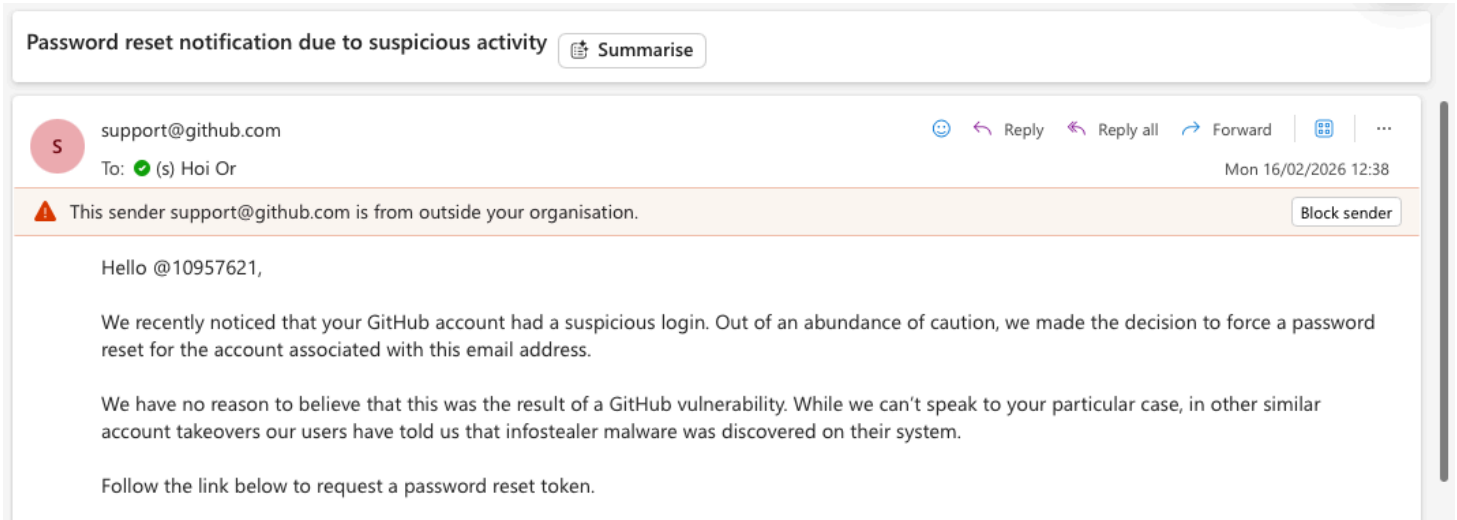


Figure C.1.1

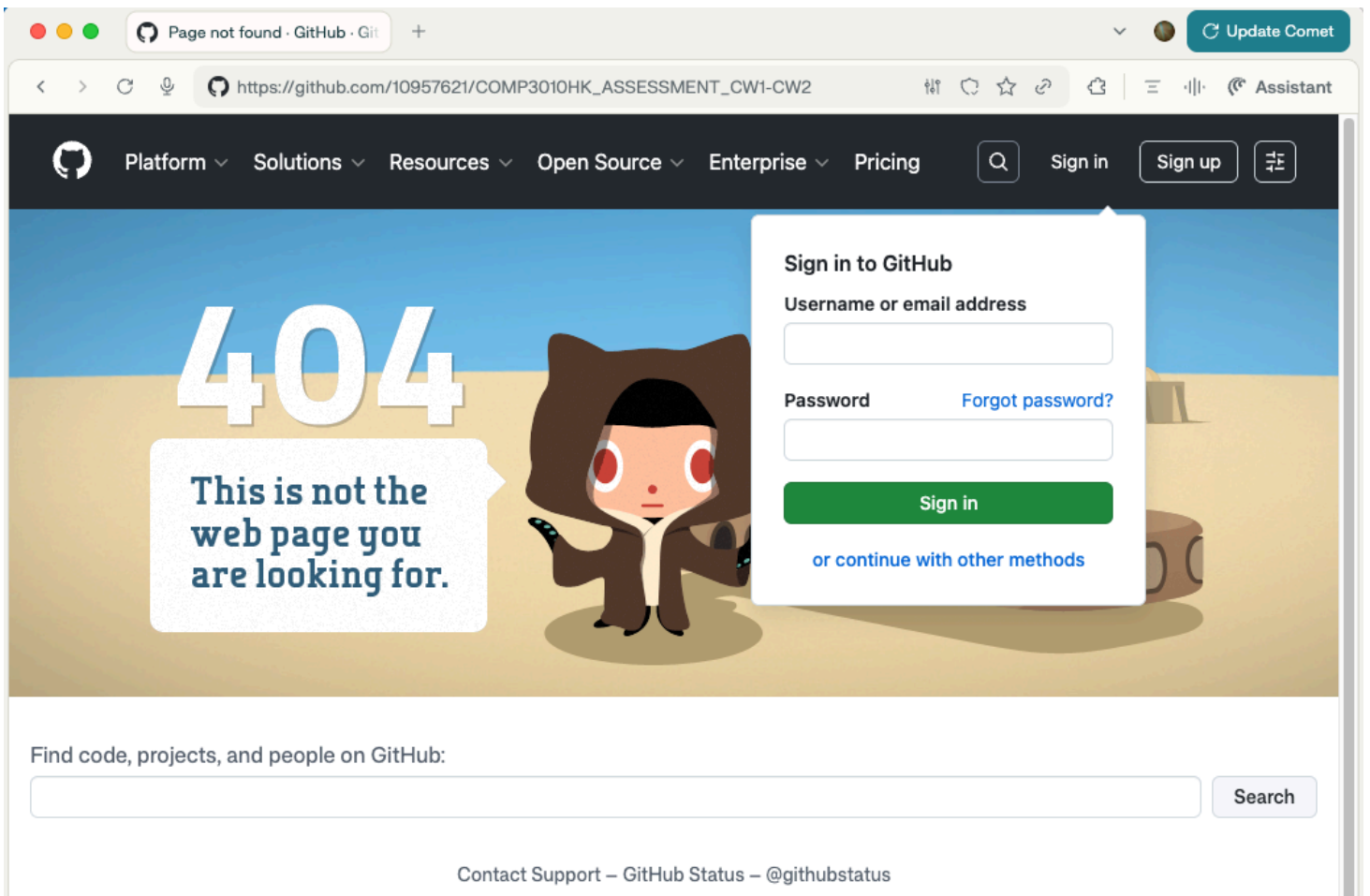


Figure C.1.2

Events matching search query



github-staff – oauth_access.destroy

Revoked a token for [10957621](#) ending in Mj3Zp6vV for the GitHub Classroom GitHub App.
4 hours ago | ...



github-staff – oauth_access.destroy

Revoked a token for [10957621](#) ending in k00GquXM for the Visual Studio Code OAuth app.
4 hours ago | ...



github-staff – oauth_authorization.destroy

Removed authorization for OAuth application due to password randomized (Git Credential Manager)
4 hours ago | ...



github-staff – oauth_access.destroy

Revoked a token for [10957621](#) ending in 9K0s8etS for the Git Credential Manager OAuth app.
4 hours ago | ...



github-staff – user.suspend

Suspended access for [10957621](#) to GitHub
4 hours ago | ...

@timestamp	2026-02-16 20:37:54 +0800
_document_id	UTWCrQjwKR6ZeRPjxMXXJQ
action	user.suspend
actor	github-staff
actor_id	126821655
created_at	2026-02-16 20:37:54 +0800
hashed_token	YJCrNdLSEa7ES3Ct8kN1wn2O4kc0bFp0MBTcn7obfZY=
operation_type	modify
programmatic_access_type	OAuth access token created before 2021-04-05
request_access_security_header	nil
request_id	7327:260E7:58F5BA:17915AE:69930FA2
token_id	427032663
token_scopes	read:enterprise,read:org,read:project,repo,site_admin,user,w...
user	10957621
user_agent	spamurai-next-graphql
user_id	185078812

Figure C.1.3

	github-staff – oauth_authorization.destroy Removed authorization for OAuth application due to password randomized (GitHub Classroom) 4 hours ago ...
	github-staff – oauth_access.destroy Revoked a token for 10957621 ending in uR3zcyVK for the GitHub Classroom GitHub App. 4 hours ago ...
	github-staff – oauth_access.destroy Revoked a token for 10957621 ending in rE2wLPvx for the GitHub Classroom GitHub App. 4 hours ago ...
	github-staff – oauth_authorization.destroy Removed authorization for OAuth application due to password randomized (Visual Studio Code) 4 hours ago ...
	github-staff – oauth_access.destroy Revoked a token for 10957621 ending in for the Visual Studio Code OAuth app. 4 hours ago ...
	github-staff – oauth_access.destroy Revoked a token for 10957621 ending in Xm0bq4uh for the Visual Studio Code OAuth app. 4 hours ago ...
	github-staff – oauth_access.destroy Revoked a token for 10957621 ending in am0M7eUE for the GitHub Classroom GitHub App. 4 hours ago ...
	github-staff – user.unsuspend Lifted 10957621 's suspension from GitHub 4 hours ago ...

Figure C.1.4

	10957621 – user.two_factor_requested Two-factor auth requested 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – user.logout Logged out 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – two_factor_authentication.add_factor An authentication factor was added 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – two_factor_authentication.enabled Enabled two-factor authentication 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – user.two_factor_recovery_codes_downloaded Two factor recovery codes downloaded 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – user.two_factor_recovery_codes_downloaded Two factor recovery codes downloaded 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – user.login Logged in 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	10957621 – user.reset_password Reset their password 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...
	GitHub System – user.forgot_password Password reset requested for 10957621 delivered to hoi.or@students.plymouth.ac.uk 4 hours ago ...
	10957621 – user.failed_login Failed to login 45.14.71.16 Osaka, Osaka, Japan 4 hours ago ...

Figure C.1.5

2. Impact on Submission and Incident Response & Verification

Then I checked the SSH and GPG keys and also reviewed the authorized applications.



10957621 Hoi-Chuen Or (10957621)

Your personal account

Go to your personal profile

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Teams

Moderation

SSH keys

New SSH key

There are no SSH keys associated with your account.

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

GPG keys

New GPG key

There are no GPG keys associated with your account.

Learn how to [generate a GPG key and add it to your account](#).

Vigilant mode

☐ Flag unsigned commits as unverified

This will include any commit attributed to your account but not signed with your GPG or S/MIME key. Note that this will include your existing unsigned commits.
[Learn about vigilant mode.](#)

Figure C.2.1



10957621 Hoi-Chuen Or (10957621)

Your personal account

Go to your personal profile

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Teams

Moderation

Applications

Installed GitHub Apps

Authorized GitHub Apps

Authorized OAuth Apps

You have granted 2 applications access to your account.

Sort

Revoke all



GitHub Desktop

Last used within the last 2 years · Owned by [desktop](#)



JetBrains IDE Integration

Last used within the last 2 years · Owned by [JetBrains](#)

Read more about connecting with third-party applications at [GitHub Help](#).

Figure C.2.2



10957621 Hoi-Chuen Or (10957621)
Your personal account

Go to your personal profile

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Teams

Moderation

Applications

Installed GitHub Apps

Authorized GitHub Apps

Authorized OAuth Apps



No authorized GitHub Apps
You have no GitHub Apps authorized to access your account.

Visit Marketplace

[My GitHub Apps](#)

Figure C.2.3

I can still upload or edit the repository, but all images I upload seem to be restricted by GitHub policy and show as damaged. Even though the repository is set as public, only I can see the content, which means it is not ready for assessment review. To fix this, I submitted a request to the GitHub Support team and explained the situation to request further assistance.

Figure C.2.5



Refresh

Return to Tickets

Account Flagged for Suspicious Activity - Security Secured (Reinstatement Request) #4089768

Open 10957621 Hoi-Chuen Or opened this ticket for 10957621 4 hours ago · 3 comments



To: 10957621 Hoi-Chuen Or GitHub Support

Write

Preview

H B I

Add a comment

Paste, drop, or click to add files



Close ticket

Comment



10957621 Hoi-Chuen Or commented 4 hours ago

Hello GitHub Support,

Thank you for the quick response. Here are the verification details for my account:

1. Account username:

10957621

2. Approximate location of where the account was created:

Hong Kong

(Note: I am currently physically located in Hong Kong, but I often use a VPN which may show my IP address as Japan. Please note this is my legitimate activity.)

3. Original email address in the account:

hoi.or@students.plymouth.ac.uk

4. Other information:

My account was flagged due to a suspicious login. Regarding the "infostealer" warning:

- Malware Check: I am using a Mac (macOS). I have inspected my installed applications and browser extensions and removed anything unused or suspicious. I have also cleared my browser cookies and active sessions to ensure no session data remains compromised.
- Password: I have successfully reset my password to a strong, unique one.
- SSH Keys: I have verified that my SSH keys list is empty (no unauthorized keys).
- 2FA: I have successfully enabled Two-Factor Authentication.

Please review my account and restore public visibility to my profile and repositories.

Thank you.

Oscar

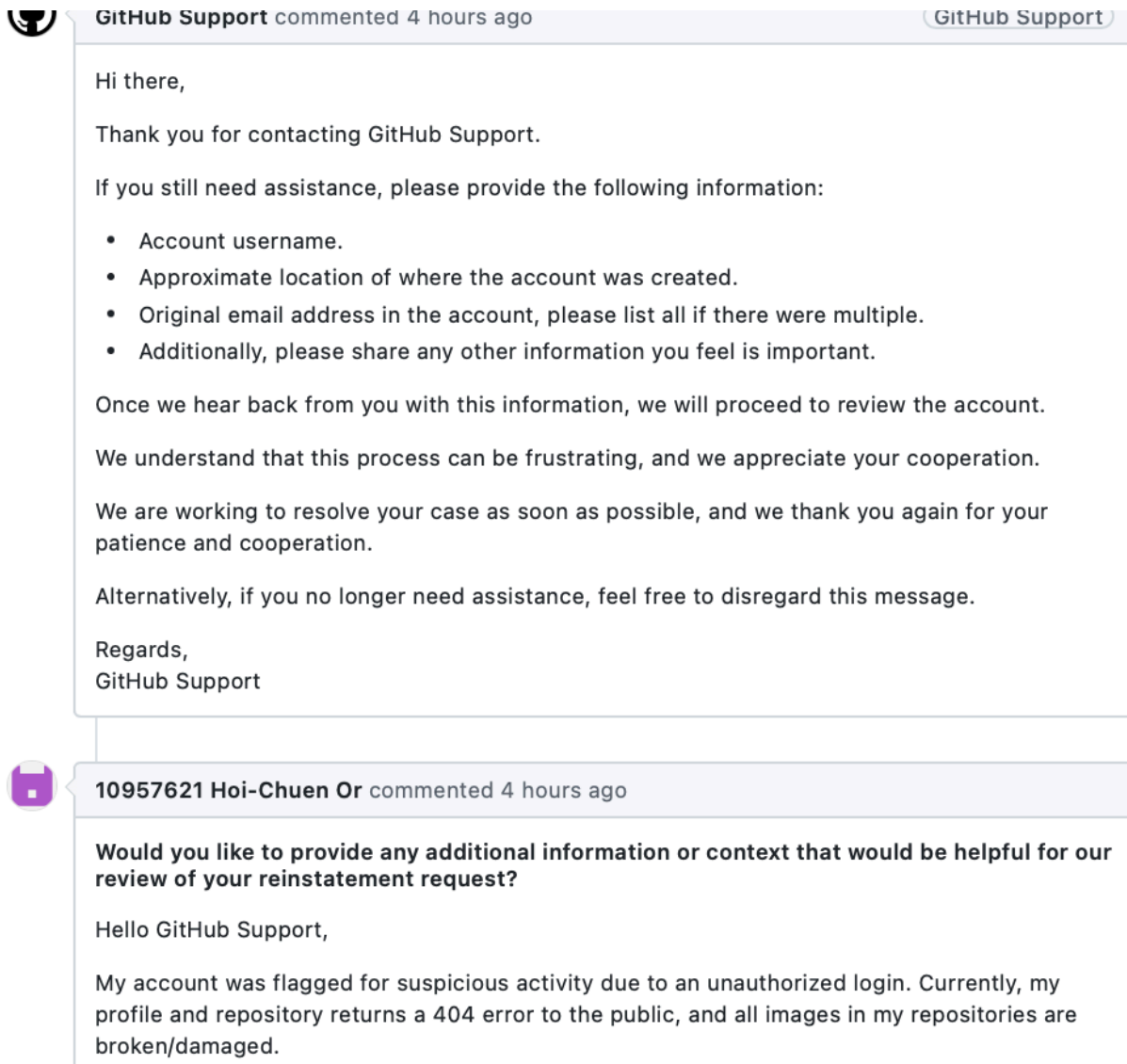


Figure C.2.6

3. Evidence of the Github work

Since my GitHub account may require a long time to complete the reinstatement request, this section presents my work for this module.

This repository contains two folders corresponding to Coursework 1 and Coursework 2.

main
1 Branch
0 Tags

Add file
<> Code

10957621 Add files via upload c613918 · 6 minutes ago 🕒 145 Commits	
CW1	Submission 3 weeks ago
CW2	Add files via upload 6 minutes ago
README.md	Initial commit last month

README

COMP3010HK_ASSESSMENT_CW1-CW2

Coursework for COMP3010HK Security Operations & Incident Management - PCAP analysis (CW1) and BOTSv3 Incident Analysis and Presentation(CW2)

Figure C.3.1

3.1 CW1

The details of CW1 follow on the next page. The evidence, such as screenshots, has already been inserted in the report and submitted in DLE previously.

Attached Documents:

- [CW1 Assessment](#)
- [CW1 Evidence](#)
- [CW1 Virus Total](#)

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2

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Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

COMP3010HK_ASSESSMENT_CW1-CW2 / CW1 / Virus Total /

Add file

10957621

Add files via upload

7214ec4 · 3 weeks ago

History

Name	Name	Last commit date
..		
.gitkeep	Create .gitkeep	3 weeks ago
C2_Server_VirusTotal - IP address ...	Rename Cobalt Strike Server_VirusTotal - ...	3 weeks ago
C2_VirusTotal - IP address - 148.72...	Rename VirusTotal - IP address - 148.72.1...	3 weeks ago
C2_VirusTotal - IP address - 148.72...	Rename VirusTotal - IP address - 148.72.5...	3 weeks ago
C2_VirusTotal - IP address - 210.24...	Rename VirusTotal - IP address - 210.245....	3 weeks ago
Cobalt Strike Server_VirusTotal - IP ...	Rename VirusTotal - IP address - 185.106....	3 weeks ago
Cobalt Strike Server_VirusTotal - IP ...	Add files via upload	3 weeks ago
VirusTotal - IP address - 10.9.23.10...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 10.9.23.5.p...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 13.107.22....	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 13.107.42.1...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 13.69.109....	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 20.190.160...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 20.49.150....	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 20.54.36.2...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 204.79.19...	VirusTotal Check result	3 weeks ago
VirusTotal - IP address - 51.105.218...	VirusTotal Check result	3 weeks ago

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2

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Code

IssuesPull requestsActionsProjectsWikiSecurityInsightsSettings

COMP3010HK_ASSESSMENT_CW1-CW2 / CW1 / Evidence /

Add file ▾

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10957621

Add files via upload

9cea370 · 3 weeks ago

History

Name	Name	Last commit date
..		
draft	Rename CW1/Evidence/Q9.2.png to CW1/...	3 weeks ago
1.1.png	Rename CW1/1.1.png to CW1/Evidence/1.1....	3 weeks ago
1.2.png	Rename CW1/1.2.png to CW1/Evidence/1.2...	3 weeks ago
2.1.png	Add files via upload	3 weeks ago
2.2.png	Rename CW1/2.2.png to CW1/Evidence/2....	3 weeks ago
2.3Zui_Alert.png	Rename CW1/2.3Zui_Alert.png to CW1/Evi...	3 weeks ago
Q1.png	Rename CW1/Q1.png to CW1/Evidence/Q1...	3 weeks ago
Q10.png	Add files via upload	3 weeks ago
Q11.png	Add files via upload	3 weeks ago
Q12.png	Add files via upload	3 weeks ago
Q13,15.png	Rename Q13-16.1.png to Q13,15.png	3 weeks ago
Q13-16.2.png	Rename CW1/Q13-16.2.png to CW1/Evide...	3 weeks ago
Q14.png	Add files via upload	3 weeks ago
Q17-20.1.png	Rename CW1/Q17-20.1.png to CW1/Evide...	3 weeks ago
Q17-20.2.png	Rename CW1/Q17-20.2.png to CW1/Evide...	3 weeks ago
COMP3010HK_ASSESSMENT_CW1-CW2 / CW1 / Evidence / ↑ Top		
Q17-20.4.png	Rename CW1/Q17-20.4.png to CW1/Evide...	3 weeks ago
Q17-20.5.png	Rename CW1/Q17-20.5.png to CW1/Evide...	3 weeks ago
Q17-20.6.png	Rename CW1/Q17-20.6.png to CW1/Evide...	3 weeks ago
Q2-Q6.1.png	Rename CW1/Q2-Q6.1.png to CW1/Eviden...	3 weeks ago
Q2-Q6.2.png	Rename CW1/Q2-Q6.2.png to CW1/Evide...	3 weeks ago
Q7.png	Rename CW1/Q7.png to CW1/Evidence/Q7...	3 weeks ago
Q8.1.png	Rename CW1/Q8.1.png to CW1/Evidence/...	3 weeks ago
Q9.1.png	Add files via upload	3 weeks ago
Q9.2.png	Add files via upload	3 weeks ago
README.MD	Update README.MD	3 weeks ago
zombie.png	Rename CW1/zombie.png to CW1/Evidenc...	3 weeks ago

https://github.com/10957621/COMP3010HK_ASSESSMENT_CW1-CW2/tree/main/CW1/Evidence

Page 1 of 2

Zui Queries

1. Victim Identification

Goal: Identify the victim's IP, MAC address, and Hostname.

Query: `_path == "dhcp" | sort ts`

2. Alert Triage

Goal: Review initial IDS alerts to establish a timeline of suspicious activity.

Query: `event_type == "alert" | sort ts | fuse`

3. Initial Infection Vector

Goal: Identify the specific HTTP GET request used to download the payload.

Query: `_path=="http" method=="GET" | sort ts | cut _path, ts, uid, id, method, host, uri`

4. Post-Infection Traffic Analysis

Goal: Analyze network traffic in the 5-minute window following the download to identify Command & Control (C2) channels.

Query: `_path=="http" or _path=="ssl" | ts >= 2021-09-24T16:44:40Z | ts <= 2021-09-24T16:49:40Z | id.orig_h == 10.9.23.102 | cut ts, id.resp_h, host, server_name, uri, resp_bytes | sort ts | fuse`

5. IP Check

Goal: Locate the external IP lookup performed by the malware before the spam campaign.

Query: `_path=="dns" query=="api.ipify.org" | sort ts`

Wireshark Filters

1. Payload Extraction

Goal: Locate and extract the malicious zip file.

Filter: `http contains "documents.zip"`

2. C2 Data Exfiltration

Goal: Filter for HTTP POST requests to identify data being sent to the attacker.

Filter: `http.request.method == "POST"`

3. C2 Beaconsing Evidence

Goal: Isolate traffic to the confirmed C2 domain (`maldivehost.net`) to observe beaconsing patterns.

Filter: `http.host == "maldivehost.net"`

4. Action on Objectives (Spam)

Goal: Analyze SMTP traffic to identify the spam campaign and stolen credentials.

Filter: `smtp`

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Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

COMP3010HK_ASSESSMENT_CW1-CW2 / CW1 /

Add file ▼

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10957621 Submission

b380bbc · 3 weeks ago

History

Name	Name	Last commit date
..		
Evidence	Add files via upload	3 weeks ago
Virus Total	Add files via upload	3 weeks ago
COMP3010HK_CW1.pdf	Submission	3 weeks ago
Conversations.xlsx	Add files via upload	3 weeks ago
README.MD	Update README.MD	3 weeks ago

README.MD

The Evidence folder stores quiz screenshot evidence and key views from the PCAP.

The PNG file names correspond to the questions; some of them are applicable to multiple questions.

The VirusTotal folder contains IP reputation checks and results from VirusTotal lookups.

The Conversations folder contains Wireshark conversation exports.

3.2 CW2

Important Note: The following Git commit records and repository evidence were captured on 2026-02-16 after the incident occurred, but the commit timestamps themselves reflect the actual dates when the work was completed (prior to the security incident). Git commit metadata is cryptographically signed and tamper-evident, ensuring the authenticity of these timestamps.

Despite the public visibility issue, I retain full access to my local repository. The following evidence demonstrates my coursework completion and authorship:

The CW2 folder contains three subfolders: quiz, Installation & Data Preparation, and Appendix_C. These folder stores the screenshots used as evidence for the CW2 report.

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2

Type to search

Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Files

main

Go to file

CW1

CW2

Appendix_C

Installation & Data Preparation

Quiz

README.MD

Student Declaration of AI Tool ...

preview.png

youtubepreview.png

README.md

COMP3010HK_ASSESSMENT_CW1-CW2 / CW2

Add file

10957621 Add files via upload c613918 · 1 hour ago History

Name	Last commit message	Last commit date
..		
Appendix_C	Add files via upload	1 hour ago
Installation & Data Preparation	Delete CW2/Installation & Data Preparation/00.jpg	last week
Quiz	Add files via upload	yesterday
README.MD	Update README.MD	1 hour ago
Student Declaration of AI Tool use in this...	Add files via upload	6 hours ago
preview.png	Add files via upload	6 hours ago
youtubepreview.png	Add files via upload	6 hours ago

Figure C.3.2.1

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2

Type to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Files

main

Go to file

> CW1

> CW2

> Appendix_C

> Installation & Data Preparati...

01.jpg

02.jpg

03.jpg

04.jpg

05.jpg

06.jpg

07.jpg

08.jpg

09.jpg

10.jpg

11.jpg

12.jpg

> Quiz

README.MD

Student Declaration of AI To...

preview.png

youtubepreview.png

README.md

COMP3010HK_ASSESSMENT_CW1-CW2 / CW2 / Installation & Data Preparation /

10957621 Delete CW2/Installation & Data Preparation/00.jpg f642f11 · last week History

Name	Last commit message	Last commit d...
..		
01.jpg	Ubuntu and Splunk	last week
02.jpg	Ubuntu and Splunk	last week
03.jpg	Ubuntu and Splunk	last week
04.jpg	Ubuntu and Splunk	last week
05.jpg	Ubuntu and Splunk	last week
06.jpg	Ubuntu and Splunk	last week
07.jpg	Ubuntu and Splunk	last week
08.jpg	Ubuntu and Splunk	last week
09.jpg	Ubuntu and Splunk	last week
10.jpg	Ubuntu and Splunk	last week
11.jpg	Ubuntu and Splunk	last week
12.jpg	Ubuntu and Splunk	last week

Figure C.3.2.2

Files

main

Q

Go to file

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CW1

>

📁

CW2

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📁

Appendix_C

>

📁

Installation & Data Preparati...

>

📁

Quiz

📄

.gitkeep

📄

Q1.jpg

📄

Q1.png

📄

Q2.1.jpg

📄

Q2.2.jpg

📄

Q2.png

📄

Q3.1.jpg

📄

Q3.2.jpg

📄

Q3.png

📄

Q4-6.png

📄

Q4.1.jpg

📄

Q5.1.jpg

📄

Q6.1.jpg

📄

Q7.1.jpg

📄

Q7.2.jpg

📄

Q7.3.jpg

📄

Q7.4.jpg

📄

Q7.png

📄

Q8.1.jpg

📄

Q8.2.jpg

📄

Q8.3.jpg

📄

Q8.png

📄

Q9.0.jpg

📄

Q9.1.jpg

📄

Q9.2.jpg

📄

Q9.3.jpg

📄

SPL7.1.png

📄

SPL7.2.png

📄

README.MD

📄

Student Declaration of AI To...

📄

preview.png

📄

youtubepreview.png

📄

README.md

COMP3010HK_ASSESSMENT_CW1-CW2 / CW2 / Quiz /

10957621

Add files via upload

171789d · yesterday

History

Name	Last commit message	Last commit d...
..		
.gitkeep	Create .gitkeep	3 weeks ago
Q1.jpg	Add files via upload	5 days ago
Q1.png	Add files via upload	3 weeks ago
Q2.1.jpg	Add files via upload	5 days ago
Q2.2.jpg	Add files via upload	5 days ago
Q2.png	Add files via upload	3 weeks ago
Q3.1.jpg	Add files via upload	4 days ago
Q3.2.jpg	Add files via upload	4 days ago
Q3.png	Add files via upload	3 weeks ago
Q4-6.png	Add files via upload	3 weeks ago
Q4.1.jpg	Add files via upload	4 days ago
Q5.1.jpg	Add files via upload	4 days ago
Q6.1.jpg	Add files via upload	4 days ago
Q7.1.jpg	Add files via upload	3 days ago
Q7.2.jpg	Add files via upload	3 days ago
Q7.3.jpg	Add files via upload	3 days ago
Q7.4.jpg	Add files via upload	3 days ago
Q7.png	Add files via upload	3 weeks ago
Q8.1.jpg	Add files via upload	3 days ago
Q8.2.jpg	Add files via upload	3 days ago
Q8.3.jpg	Add files via upload	3 days ago
Q8.png	Add files via upload	3 weeks ago
Q9.0.jpg	Add files via upload	2 days ago
Q9.1.jpg	Add files via upload	2 days ago
Q9.2.jpg	Add files via upload	2 days ago
Q9.3.jpg	Add files via upload	2 days ago
SPL7.1.png	Add files via upload	yesterday
SPL7.2.png	Add files via upload	yesterday

Figure C.3.2.3

3.3 CW2 Commits History

Commits

main	All users	All time
Commits on Feb 17, 2026		
Add files via upload 10957621 authored 35 minutes ago Verifiedc613918 <> Create .gitkeep 10957621 authored 36 minutes ago Verified8a250c4 <> Update README.MD 10957621 authored 37 minutes ago Verified4394543 <>		
Commits on Feb 16, 2026		
Add files via upload 10957621 authored 5 hours ago Verifiedfb0ba2b <> Delete CW2/youtubepreview.png 10957621 authored 5 hours ago Verified7919b40 <> Delete CW2/preview.png 10957621 authored 5 hours ago Verified5ae5b04 <> Update README.MD 10957621 authored 5 hours ago Verifiedca0e964 <> Update README.MD 10957621 authored 6 hours ago Verified5de75bb <> Update README.MD 10957621 authored 6 hours ago Verifiedad88368 <> Update README.MD 10957621 authored 6 hours ago Verified899670d <> Update README.MD 10957621 authored 6 hours ago Verified31c3619 <> Update README.MD 10957621 authored 6 hours ago Verifieda4169a2 <> Add files via upload 10957621 authored 6 hours ago Verified7f921c3 <> Add files via upload 10957621 authored 6 hours ago Verified227b646 <> Update README.MD 10957621 authored 6 hours ago Verified7c6025d <>		
Commits on Feb 15, 2026		
Add files via upload 10957621 authored yesterday Verified171789d <> Update README.MD 10957621 authored yesterday Verified49afc15 <> Update README.MD 10957621 authored yesterday Verified7ba7a38 <> Add files via upload 10957621 authored yesterday Verifiedb79e6f1 <> Update README.MD Verifiedfa9a73e <>		

10957621 authored yesterday			
Update README.MD	10957621 authored yesterday	Verified c4388de	<>
Update README.MD	10957621 authored yesterday	Verified d0649b9	<>
Update README.MD	10957621 authored yesterday	Verified 1e4585e	<>
Update README.MD	10957621 authored yesterday	Verified 5a446b1	<>
Update README.MD	10957621 authored yesterday	Verified f34ad51	<>
Update README.MD	10957621 authored yesterday	Verified 65d5e50	<>

Commits on Feb 14, 2026

Update README.MD	10957621 authored 2 days ago	Verified d8b7e12	<>
Update README.MD	10957621 authored 2 days ago	Verified cfb045a	<>
Update README.MD	10957621 authored 2 days ago	Verified 45c4911	<>
Update README.MD	10957621 authored 2 days ago	Verified be0b204	<>
Update README.MD	10957621 authored 2 days ago	Verified 66680f2	<>
Update README.MD	10957621 authored 2 days ago	Verified 8252eb8	<>
Update README.MD	10957621 authored 2 days ago	Verified 4eb4a9b	<>
Add files via upload	10957621 authored 2 days ago	Verified 192e6a3	<>
Update section Q8	10957621 authored 3 days ago	Verified 1ca7d9e	<>

[Previous](#)
[Next](#)
[>](#)

Figure 3.3.1

Commits

🔗 main	🔍 All users	📅 All time
Commits on Feb 14, 2026		
Add files via upload 👤 10957621 authored 3 days ago Verified35a5c37📄 <>		
Add files via upload 👤 10957621 authored 3 days ago Verified6bb04d8📄 <>		
Update section4 Q7 👤 10957621 authored 3 days ago Verified5c4f56d📄 <>		
Commits on Feb 13, 2026		
Update section4 Q4-6 👤 10957621 authored 4 days ago Verified0016326📄 <>		
Update section 4 Q4-6 👤 10957621 authored 4 days ago Verified0918d01📄 <>		
Commits on Feb 12, 2026		
Add files via upload 👤 10957621 authored 4 days ago Verifieda26dbe0📄 <>		
Update section4 Q3 👤 10957621 authored 4 days ago Verifiedd07668c📄 <>		
Add files via upload 👤 10957621 authored 4 days ago Verified3ee8cd6📄 <>		
Update README.MD 👤 10957621 authored 5 days ago Verified5d76cf1📄 <>		
Update README.MD 👤 10957621 authored 5 days ago Verified37fe614📄 <>		
Update section4 Q2 👤 10957621 authored 5 days ago Verifiede0051a7📄 <>		
Update section 4 Q2 👤 10957621 authored 5 days ago Verified45a9e02📄 <>		
Add files via upload 👤 10957621 authored 5 days ago Verifiedf01547d📄 <>		
Update section 4 q2 👤 10957621 authored 5 days ago Verified531af77📄 <>		
Commits on Feb 11, 2026		
Update section4 q1 👤 10957621 authored 5 days ago Verified2ff9fb6📄 <>		
Add files via upload 👤 10957621 authored 5 days ago Verified309e45e📄 <>		
Update README.MD 👤 10957621 authored 5 days ago Verifiedcc75fa0📄 <>		
Update section 4 👤 10957621 authored 5 days ago Verified7a506f0📄 <>		
Commits on Feb 9, 2026		

Update section 2 & 3 10957621 authored last week	Verified	ba5aece		<>
Update section3 10957621 authored last week	Verified	026954c		<>
Update section3 10957621 authored last week	Verified	eab249b		<>
Update section3 10957621 authored last week	Verified	d30076e		<>
Update section2 10957621 authored last week	Verified	f5ea96f		<>
Update section3 cap screen 10957621 authored last week	Verified	eabc788		<>
Update README.MD 10957621 authored last week	Verified	5f77ebc		<>
Update README.MD 10957621 authored last week	Verified	e8539f5		<>
Update section3 10957621 authored last week	Verified	c6bd76e		<>
section3 10957621 authored last week	Verified	e5110ad		<>
Delete CW2/Installation & Data Preparation/00.jpg 10957621 authored last week	Verified	f642f11		<>

Commits on Feb 8, 2026

Ubuntu and Splunk 10957621 authored last week	Verified	ffb6749		<>
Delete CW2/Installation & Data Preparation/.gitkeep 10957621 authored last week	Verified	31f3973		<>
Ubuntu and Splunk 10957621 authored last week	Verified	f215c40		<>
Create .gitkeep 10957621 authored last week	Verified	51faca4		<>
Update README.MD 10957621 authored last week	Verified	51267e3		<>

Commits on Feb 7, 2026

Update README.MD 10957621 authored last week	Verified	93b7b34		<>
---	----------	---------	--	----

< Previous Next >

Figure 3.3.2

Commits

main	All users	All time
Commits on Feb 7, 2026		
Update README.MD 10957621 authored last week Verified303e970		
Commits on Feb 5, 2026		
Update README.MD 10957621 authored 2 weeks ago Verified4b25d00		
Commits on Jan 29, 2026		
Submission 10957621 authored 3 weeks ago Verifiedb380bbc		
Add files via upload 10957621 authored 3 weeks ago Verified02e1f1d		
Add files via upload 10957621 authored 3 weeks ago Verified9cea370		
Rename CW1/Evidence/Q9.2.png to CW1/Evidence/draft/Q9.2.png 10957621 authored 3 weeks ago Verifiedbe16ce0		
Rename CW1/Evidence/Q9.1.png to CW1/Evidence/draft/Q9.1.png 10957621 authored 3 weeks ago Verified7cbb3da		
Rename CW1/Evidence/Q12.png to CW1/Evidence/draft/Q12.png 10957621 authored 3 weeks ago Verified4c4299a		
Rename CW1/Evidence/Q10-11.png to CW1/Evidence/draft/Q10-11.png 10957621 authored 3 weeks ago Verifiedfc390dc		
Create .gitkeep 10957621 authored 3 weeks ago Verified69e8538		
Add files via upload 10957621 authored 3 weeks ago Verified7214ec4		
Rename Cobalt Strike Server_VirusTotal - IP address - 136.232.34.70.pdf to C2_Server_VirusTotal - IP address - 136.232.34.70.pdf 10957621 authored 3 weeks ago Verified6059c64		
Rename VirusTotal - IP address - 210.245.90.247.pdf to C2_VirusTotal - IP address - 210.245.90.247.pdf 10957621 authored 3 weeks ago Verified76b886b		
Rename VirusTotal - IP address - 148.72.53.144.pdf to C2_VirusTotal - IP address - 148.72.53.144.pdf 10957621 authored 3 weeks ago Verifiede82538b		
Rename VirusTotal - IP address - 148.72.192.206.pdf to C2_VirusTotal - IP address - 148.72.192.206.pdf 10957621 authored 3 weeks ago Verified9a4fed3		
Rename VirusTotal - IP address - 136.232.34.70.pdf to Cobalt Strike Server_VirusTotal - IP address - 136.232.34.70.pdf 10957621 authored 3 weeks ago Verified59d768b		
Rename VirusTotal - IP address - 185.106.96.158.pdf to Cobalt Strike Server_VirusTotal - IP address - 185.106.96.158.pdf 10957621 authored 3 weeks ago Verified5b6e9a1		
Add files via upload 10957621 authored 3 weeks ago Verified93f9ed5		
Add files via upload		

 10957621 authored 3 weeks ago	verified	94a417b		↔
Delete CW1/Evidence/Q10-12.3.png  10957621 authored 3 weeks ago	Verified	0c00656		↔
Delete CW1/Evidence/Q10-12.2.png  10957621 authored 3 weeks ago	Verified	4c91e8b		↔
Delete CW1/Evidence/Q10-12.1.png  10957621 authored 3 weeks ago	Verified	538f7fd		↔
Update README.MD  10957621 authored 3 weeks ago	Verified	078a52f		↔
Update README.MD  10957621 authored 3 weeks ago	Verified	17f35c1		↔
Update README.MD  10957621 authored 3 weeks ago	Verified	e93bdf6		↔
Update README.MD  10957621 authored 3 weeks ago	Verified	37e60cf		↔
Create README.MD  10957621 authored 3 weeks ago	Verified	045a957		↔

-
Commits on Jan 28, 2026

Add files via upload  10957621 authored 3 weeks ago	Verified	0d59a1f		↔
Create .gitkeep  10957621 authored 3 weeks ago	Verified	badb0af		↔
Add files via upload  10957621 authored 3 weeks ago	Verified	84e918e		↔
Delete CW1/Evidence/Q9.2.png  10957621 authored 3 weeks ago	Verified	d44aa9c		↔
Add files via upload  10957621 authored 3 weeks ago	Verified	fa148ba		↔
Rename Q13-16.1.png to Q13,15.png  10957621 authored 3 weeks ago	Verified	9ddf431		↔

-
Commits on Jan 27, 2026

Update README.MD  10957621 authored 3 weeks ago	Verified	d73fe97		↔
Update README.MD  10957621 authored 3 weeks ago	Verified	e8c2f1e		↔

[< Previous](#)
[Next >](#)

Figure 3.3.3

Commits

main

All users

All time

Commits on Jan 27, 2026

Conversations	Verified	de5a026		<>
 10957621 authored 3 weeks ago				
Rename CW1/zombie.png to CW1/Evidence/zombie.png	Verified	152cc18		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q9.2.png to CW1/Evidence/Q9.2.png	Verified	63a2dba		<>
 10957621 authored 3 weeks ago				
Delete CW1/Evidence/Q8.2.png	Verified	c2d493e		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q9.1.png to CW1/Evidence/Q9.1.png	Verified	6360604		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q8.2.png to CW1/Evidence/Q8.2.png	Verified	97d8922		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q8.1.png to CW1/Evidence/Q8.1.png	Verified	0923cd1		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q7.png to CW1/Evidence/Q7.png	Verified	b83e8e4		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q2-Q6.2.png to CW1/Evidence/Q2-Q6.2.png	Verified	5605cf8		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q2-Q6.1.png to CW1/Evidence/Q2-Q6.1.png	Verified	2ee76b5		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.6.png to CW1/Evidence/Q17-20.6.png	Verified	4bc3a3a		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.5.png to CW1/Evidence/Q17-20.5.png	Verified	8f4f8b5		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.4.png to CW1/Evidence/Q17-20.4.png	Verified	e763894		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.3.png to CW1/Evidence/Q17-20.3.png	Verified	d4c7077		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.2.png to CW1/Evidence/Q17-20.2.png	Verified	2d66a8f		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q17-20.1.png to CW1/Evidence/Q17-20.1.png	Verified	81440be		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q13-16.2.png to CW1/Evidence/Q13-16.2.png	Verified	c324aad		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q13-16.1.png to CW1/Evidence/Q13-16.1.png	Verified	f6ee1b1		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q10-12.3.png to CW1/Evidence/Q10-12.3.png	Verified	9810156		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q10-12.2.png to CW1/Evidence/Q10-12.2.png	Verified	93be643		<>
 10957621 authored 3 weeks ago				
Rename CW1/Q10-12.1.png to CW1/Evidence/Q10-12.1.png	Verified	f4bb944		<>
 10957621 authored 3 weeks ago				

Rename CW1/Q1.png to CW1/Evidence/Q1.png 10957621 authored 3 weeks ago	Verified	ccaccf5		<>
Rename CW1/2.3Zui_Alert.png to CW1/Evidence/2.3Zui_Alert.png 10957621 authored 3 weeks ago	Verified	8be747d		<>
Rename CW1/2.2.png to CW1/Evidence/2.2.png 10957621 authored 3 weeks ago	Verified	a2868b2		<>
Add files via upload 10957621 authored 3 weeks ago	Verified	8323f4d		<>
Delete CW1/Evidence /Evidence directory 10957621 authored 3 weeks ago	Verified	5d971fe		<>
Rename CW1/2.1.png to CW1/Evidence /Evidence/2.1.png 10957621 authored 3 weeks ago	Verified	ae4529a		<>
Rename CW1/1.2.png to CW1/Evidence/1.2.png 10957621 authored 3 weeks ago	Verified	77e3426		<>
Delete CW1/Evidence/.gitkeep 10957621 authored 3 weeks ago	Verified	18e0c30		<>
Rename CW1/1.1.png to CW1/Evidence/1.1.png 10957621 authored 3 weeks ago	Verified	4344f31		<>
Create .gitkeep 10957621 authored 3 weeks ago	Verified	bf41dd8		<>
VirusTotal Check result 10957621 authored 3 weeks ago	Verified	b60c6af		<>
Create .gitkeep 10957621 authored 3 weeks ago	Verified	3015f4b		<>
Delete CW1/Virus Total 10957621 authored 3 weeks ago	Verified	7205cab		<>
Create Virus Total 10957621 authored 3 weeks ago	Verified	c6092cc		<>

< Previous Next >

Figure 3.3.4

10957621 / COMP3010HK_ASSESSMENT_CW1-CW2

Q Type to search

<> Code

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Pull requests

Actions

Projects

Wiki

Security

Insights

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Commits

main

All users

All time

Commits on Jan 27, 2026

Add files via upload

10957621 authored 3 weeks ago

Verified0404071

Evidence

10957621 authored 3 weeks ago

Verified2cd1445

Commits on Jan 6, 2026

Create README.MD

10957621 authored on Jan 6

Verified3fa75d9

Create README.MD

10957621 authored on Jan 6

Verified95c5f24

Initial commit

10957621 authored on Jan 6

Verifiedbe374b6

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Figure 3.3.5